

rph8-g15q: formula evidence

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Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
rph8-g15q_F00001	1	■ 0.280	{ }^{\text{@}}	@	Arthur J. Parzygnat ^{1,2,*} Tai-Danae Bradley ^{3,4,†} Andrew Vlasic ^{5,‡} and Anh Pham ⁶
rph8-g15q_F00002	1	■ 0.280	{ }^{1,2,*}	1,2,*	Arthur J. Parzygnat ^{1,2,*} Tai-Danae Bradley ^{3,4,†} Andrew Vlasic ^{5,‡} and Anh Pham ⁶
rph8-g15q_F00003	1	■ 0.280	{ }^{3,4,\dag}	3,4,†	Arthur J. Parzygnat ^{1,2,*} Tai-Danae Bradley ^{3,4,†} Andrew Vlasic ^{5,‡} and Anh Pham ⁶
rph8-g15q_F00004	1	■ 0.280	{ }^{5,\ddag}	5,‡	Arthur J. Parzygnat ^{1,2,*} Tai-Danae Bradley ^{3,4,†} Andrew Vlasic ^{5,‡} and Anh Pham ⁶
rph8-g15q_F00005	1	■ 1.000	{ }^{\dag}	†	*Contact author: tai.danae@math3ma.com
rph8-g15q_F00006	1	■ 1.000	{ }^{\ddag}	‡	*Contact author: avlasic@deloitte.com
rph8-g15q_F00007	3	■ 1.000	x	x	transfers classical data x onto the quantum computer, the variational
rph8-g15q_F00008	3	■ 1.000	\theta	θ	block whose parameters θ can be modified so as to optimize some
rph8-g15q_F00009	3	■ 0.466	\rangle	$\rangle$	1011 is assigned to the four-qubit state $ 1011\rangle$. Meanwhile,
rph8-g15q_F00010	3	■ 1.000	v=\left(v_{1}, \ldots, v_{d}\right)	$v = (v_1, \dots, v_d)$	encoding vectors $v = (v_1, \dots, v_d)$ in $\mathbb{R}^d$ into quantum states
rph8-g15q_F00011	3	■ 1.000	\mathbb{R}^d	$\mathbb{R}^d$	encoding vectors $v = (v_1, \dots, v_d)$ in $\mathbb{R}^d$ into quantum states
rph8-g15q_F00012	3	■ 1.000	\mathcal{H}	$\mathcal{H}$	[8,60,61]. Given a Hilbert space $\mathcal{H}$, let $\mathcal{S}(\mathcal{H})$ denote the set
rph8-g15q_F00013	3	■ 1.000	\mathscr{S}(\mathcal{H})	$\mathcal{S}(\mathcal{H})$	[8,60,61]. Given a Hilbert space $\mathcal{H}$, let $\mathcal{S}(\mathcal{H})$ denote the set
rph8-g15q_F00014	3	■ 1.000	\mathscr{U}(\mathcal{H})	$\mathcal{U}(\mathcal{H})$	let $\mathcal{U}(\mathcal{H})$ denote the set of unitary operators on $\mathcal{H}$. Although
rph8-g15q_F00015	3	■ 1.000	\mathcal{X}	$\mathcal{X}$	We will often let $\mathcal{X}$ denote a set, called a <i>data domain</i> or
rph8-g15q_F00016	3	■ 1.000	X	X	X typically used for the dataset itself (e.g., a sample), which
rph8-g15q_F00017	3	■ 1.000	\rho: \mathcal{X} \rightarrow \mathscr{S}(\mathcal{H})	$\rho: \mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$	main $\mathcal{X}$ into a Hilbert space $\mathcal{H}$ is a function $\rho: \mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$.
rph8-g15q_F00018	3	■ 1.000	U: \mathcal{X} \rightarrow \mathscr{U}(\mathcal{H})	$U: \mathcal{X} \rightarrow \mathcal{U}(\mathcal{H})$	A <i>quantum unitary encoding</i> is a function $U: \mathcal{X} \rightarrow \mathcal{U}(\mathcal{H})$.
rph8-g15q_F00019	3	■ 1.000	\psi_0\rangle \in \mathcal{H}	$ \psi_0\rangle \in \mathcal{H}$	$ \psi_0\rangle \in \mathcal{H}$, gives rise to a quantum state encoding $\rho: \mathcal{X} \rightarrow$
rph8-g15q_F00020	3	■ 1.000	\rho: \mathcal{X} \rightarrow	$\rho: \mathcal{X} \rightarrow$	$ \psi_0\rangle \in \mathcal{H}$, gives rise to a quantum state encoding $\rho: \mathcal{X} \rightarrow$
rph8-g15q_F00021	4	■ 1.000	G	G	are equipped with an action of a group G , and a quantum state
rph8-g15q_F00022	4	■ 1.000	\alpha: G \rightarrow \operatorname{Aut}(\mathcal{X})	$\alpha: G \rightarrow \operatorname{Aut}(\mathcal{X})$	homomorphism $\alpha: G \rightarrow \operatorname{Aut}(\mathcal{X})$, where $\operatorname{Aut}(\mathcal{X})$ denotes the
rph8-g15q_F00023	4	■ 1.000	\operatorname{Aut}(\mathcal{X})	$\operatorname{Aut}(\mathcal{X})$	homomorphism $\alpha: G \rightarrow \operatorname{Aut}(\mathcal{X})$, where $\operatorname{Aut}(\mathcal{X})$ denotes the
rph8-g15q_F00024	4	■ 1.000	\alpha	α	map α is referred to as the <i>group action</i> . Such a G -set $\mathcal{X}$ is
rph8-g15q_F00025	4	■ 1.000	(\mathcal{X}, \alpha)	$(\mathcal{X}, \alpha)$	often written as a pair $(\mathcal{X}, \alpha)$.
rph8-g15q_F00026	4	■ 1.000	g \in G	$g \in G$	with a group element $g \in G$ is often written as $\alpha_g: \mathcal{X} \rightarrow \mathcal{X}$.
rph8-g15q_F00027	4	■ 1.000	\alpha_g: \mathcal{X} \rightarrow \mathcal{X}	$\alpha_g: \mathcal{X} \rightarrow \mathcal{X}$	with a group element $g \in G$ is often written as $\alpha_g: \mathcal{X} \rightarrow \mathcal{X}$.
rph8-g15q_F00028	4	■ 1.000	x \in \mathcal{X}	$x \in \mathcal{X}$	Its action on an element $x \in \mathcal{X}$ is written as $\alpha_g(x)$, which is
rph8-g15q_F00029	4	■ 1.000	\alpha_g(x)	$\alpha_g(x)$	Its action on an element $x \in \mathcal{X}$ is written as $\alpha_g(x)$, which is
rph8-g15q_F00030	4	■ 1.000	\alpha_{gh}(x) = \alpha_g(\alpha_h(x))	$\alpha_{gh}(x) = \alpha_g(\alpha_h(x))$	in that $\alpha_{gh}(x) = \alpha_g(\alpha_h(x))$ for all $g, h \in G$ and $x \in \mathcal{X}$ as well
rph8-g15q_F00031	4	■ 1.000	g, h \in G	$g, h \in G$	in that $\alpha_{gh}(x) = \alpha_g(\alpha_h(x))$ for all $g, h \in G$ and $x \in \mathcal{X}$ as well
rph8-g15q_F00032	4	■ 0.987	\alpha_{1_G} = \operatorname{id}_{\mathcal{X}}	$\alpha_{1_G} = \operatorname{id}_{\mathcal{X}}$	as $\alpha_{1_G} = \operatorname{id}_{\mathcal{X}}$, where 1_G is the identity in the group G .
rph8-g15q_F00033	4	■ 0.987	1_G	1_G	as $\alpha_{1_G} = \operatorname{id}_{\mathcal{X}}$, where 1_G is the identity in the group G .
rph8-g15q_F00034	4	■ 1.000	V: G \rightarrow	$V: G \rightarrow$	As an example, if $\mathcal{H}$ is a Hilbert space, and if $V: G \rightarrow$
rph8-g15q_F00035	4	■ 1.000	\operatorname{Aut}(\mathcal{H})	$\operatorname{Aut}(\mathcal{H})$	viewed as a subset of $\operatorname{Aut}(\mathcal{H})$, since unitary operators can be
rph8-g15q_F00036	4	■ 1.000	\operatorname{Ad}_{V_g} \in \operatorname{Aut}(\mathscr{S}(\mathcal{H}))	$\operatorname{Ad}_{V_g} \in \operatorname{Aut}(\mathcal{S}(\mathcal{H}))$	operator $\operatorname{Ad}_{V_g} \in \operatorname{Aut}(\mathcal{S}(\mathcal{H}))$ defined by
rph8-g15q_F00037	4	■ 1.000	\alpha_{(1,0)}	$\alpha_{(1,0)}$	binary classification task with discrete symmetries given by $\alpha_{(1,0)}$,
rph8-g15q_F00038	4	■ 1.000	x_2 = x_1, \alpha_{(0,1)}	$x_2 = x_1, \alpha_{(0,1)}$	a reflection along the line $x_2 = x_1$, $\alpha_{(0,1)}$, an inversion, and $\alpha_{(1,1)}$,
rph8-g15q_F00039	4	■ 1.000	\alpha_{(1,1)}	$\alpha_{(1,1)}$	a reflection along the line $x_2 = x_1$, $\alpha_{(0,1)}$, an inversion, and $\alpha_{(1,1)}$,
rph8-g15q_F00040	4	■ 1.000	x_2 = -x_1	$x_2 = -x_1$	a reflection along the line $x_2 = -x_1$. The structure is periodic so
rph8-g15q_F00041	4	■ 1.000	\sigma \in \mathscr{S}(\mathcal{H})	$\sigma \in \mathcal{S}(\mathcal{H})$	for all $\sigma \in \mathcal{S}(\mathcal{H})$. Here, “Ad” stands for the <i>adjoint</i> action.
rph8-g15q_F00042	4	■ 0.987	\left(\mathscr{S}(\mathcal{H}), \operatorname{Ad}_V\right)	$(\mathcal{S}(\mathcal{H}), \operatorname{Ad}_V)$	Thus, $(\mathcal{S}(\mathcal{H}), \operatorname{Ad}_V)$ is a G -set as well.

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rph8-g15q_F00043	4	■ 1.000	$V: G \rightarrow \mathcal{U}(\mathcal{H})$	$V: G \rightarrow \mathcal{U}(\mathcal{H})$	tation $V: G \rightarrow \mathcal{U}(\mathcal{H})$ of G on the Hilbert space $\mathcal{H}$ such that
rph8-g15q_F00044	4	■ 1.000	ρ	ρ	ρ is G -equivariant, that is,
rph8-g15q_F00045	4	■ 0.616	$\mathcal{X}, \alpha$	$\mathcal{X}, \alpha$	<i>Definition 4.</i> Let $(\mathcal{X}, \alpha)$ and $(\mathcal{Y}, \beta)$ be two G -sets. A G -
rph8-g15q_F00046	4	■ 0.616	$\mathcal{Y}, \beta$	$\mathcal{Y}, \beta$	<i>Definition 4.</i> Let $(\mathcal{X}, \alpha)$ and $(\mathcal{Y}, \beta)$ be two G -sets. A G -
rph8-g15q_F00047	4	■ 0.999	$(\mathcal{Y}, \beta)$	$(\mathcal{Y}, \beta)$	equivariant map from $(\mathcal{X}, \alpha)$ to $(\mathcal{Y}, \beta)$ is a function $f: \mathcal{X} \rightarrow$
rph8-g15q_F00048	4	■ 0.999	$f: \mathcal{X} \rightarrow$	$f: \mathcal{X} \rightarrow$	equivariant map from $(\mathcal{X}, \alpha)$ to $(\mathcal{Y}, \beta)$ is a function $f: \mathcal{X} \rightarrow$
rph8-g15q_F00049	4	■ 1.000	$\mathcal{Y}$	$\mathcal{Y}$	$\mathcal{Y}$ such that
rph8-g15q_F00050	5	■ 1.000	$\mathcal{X} = \mathbb{R}^2$	$\mathcal{X} = \mathbb{R}^2$	two classes of points within a data domain $\mathcal{X} = \mathbb{R}^2$ as shown
rph8-g15q_F00051	5	■ 1.000	$y: \mathcal{X} \rightarrow \mathbb{R}$	$y: \mathcal{X} \rightarrow \mathbb{R}$	where $y: \mathcal{X} \rightarrow \mathbb{R}$ is a function that determines the class of
rph8-g15q_F00052	5	■ 0.998	$y(x) > 0$	$y(x) > 0$	any input data point via $y(x) > 0$ implies x is of class $+1$,
rph8-g15q_F00053	5	■ 0.516	$y(x) < 0$	$y(x) < 0$	while $y(x) < 0$ implies x is of class -1 . The symmetry de-
rph8-g15q_F00054	5	■ 1.000	$G \rightarrow \text{Aut}(\mathbb{R}^2)$	$G \rightarrow \text{Aut}(\mathbb{R}^2)$	$G \rightarrow \text{Aut}(\mathbb{R}^2)$ on $\mathcal{X} = \mathbb{R}^2$ given by the linear transformations
rph8-g15q_F00055	5	■ 1.000	$\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$	$\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$	Now set $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$ and set $V: G \rightarrow \mathcal{U}(\mathcal{H})$ to be the
rph8-g15q_F00056	5	■ 0.998	$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \mathbb{1}_2$	$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \mathbb{1}_2$	where X is the Pauli matrix $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\mathbb{1}_2$ is the 2×2
rph8-g15q_F00057	5	■ 0.998	2×2	2×2	where X is the Pauli matrix $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\mathbb{1}_2$ is the 2×2
rph8-g15q_F00058	5	■ 0.994	$\text{SWAP}(\psi\rangle \otimes \phi\rangle) = \phi\rangle \otimes \psi\rangle$	$\text{SWAP}(\psi\rangle \otimes \phi\rangle) = \phi\rangle \otimes \psi\rangle$	by $\text{SWAP}(\psi\rangle \otimes \phi\rangle) = \phi\rangle \otimes \psi\rangle$ for all $ \psi\rangle, \phi\rangle \in \mathbb{C}^2$.
rph8-g15q_F00059	5	■ 0.994	$ \psi\rangle, \phi\rangle \in \mathbb{C}^2$	$ \psi\rangle, \phi\rangle \in \mathbb{C}^2$	by $\text{SWAP}(\psi\rangle \otimes \phi\rangle) = \phi\rangle \otimes \psi\rangle$ for all $ \psi\rangle, \phi\rangle \in \mathbb{C}^2$.
rph8-g15q_F00060	5	■ 1.000	$V_g \rho(x) V_g^\dagger = \rho(\alpha_g(x))$	$V_g \rho(x) V_g^\dagger = \rho(\alpha_g(x))$	be a map $\rho: \mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$ such that $V_g \rho(x) V_g^\dagger = \rho(\alpha_g(x))$ for
rph8-g15q_F00061	5	■ 1.000	$x = (x_1, x_2) \in \mathcal{X}$	$x = (x_1, x_2) \in \mathcal{X}$	sends a point $x = (x_1, x_2) \in \mathcal{X}$ to
rph8-g15q_F00062	5	■ 1.000	Z	Z	where Z is the Pauli matrix $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. From this unitary
rph8-g15q_F00063	5	■ 1.000	$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	where Z is the Pauli matrix $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. From this unitary
rph8-g15q_F00064	5	■ 1.000	$0 < p < 1$	$0 < p < 1$	where $0 < p < 1$. (The value of p chosen to produce Fig. 3
rph8-g15q_F00065	5	■ 1.000	p	p	where $0 < p < 1$. (The value of p chosen to produce Fig. 3
rph8-g15q_F00066	5	■ 1.000	$ \pm, \pm\rangle := \pm\rangle \otimes \pm\rangle$	$ \pm, \pm\rangle := \pm\rangle \otimes \pm\rangle$	and $ \pm, \pm\rangle := \pm\rangle \otimes \pm\rangle$. Thus, the resulting state encoding
rph8-g15q_F00067	5	■ 0.994	$x = (x_1, x_2)$	$x = (x_1, x_2)$	sends $x = (x_1, x_2)$ to
rph8-g15q_F00068	5	■ 1.000	$\mathcal{L}: \mathbb{R}^2 \rightarrow \mathbb{M}_4$	$\mathcal{L}: \mathbb{R}^2 \rightarrow \mathbb{M}_4$	where $\mathcal{L}: \mathbb{R}^2 \rightarrow \mathbb{M}_4$ is a linear transformation sending each
rph8-g15q_F00069	5	■ 1.000	$\mathcal{L}(x) \in \mathbb{M}_4$	$\mathcal{L}(x) \in \mathbb{M}_4$	point $x \in \mathcal{X}$ to a Hermitian operator $\mathcal{L}(x) \in \mathbb{M}_4$, where $\mathbb{M}_4$
rph8-g15q_F00070	5	■ 1.000	$\mathbb{M}_4$	$\mathbb{M}_4$	point $x \in \mathcal{X}$ to a Hermitian operator $\mathcal{L}(x) \in \mathbb{M}_4$, where $\mathbb{M}_4$
rph8-g15q_F00071	5	■ 0.968	$-i\mathcal{L}(x) \in \mathfrak{su}(4)$	$-i\mathcal{L}(x) \in \mathfrak{su}(4)$	that $-i\mathcal{L}(x) \in \mathfrak{su}(4)$, the Lie algebra of 4×4 special unitary
rph8-g15q_F00072	5	■ 0.968	4×4	4×4	that $-i\mathcal{L}(x) \in \mathfrak{su}(4)$, the Lie algebra of 4×4 special unitary
rph8-g15q_F00073	5	■ 1.000	$\mathcal{L}$	$\mathcal{L}$	matrices. Such a linear transformation $\mathcal{L}$ is uniquely deter-
rph8-g15q_F00074	5	■ 1.000	$e_1 = (1, 0)$	$e_1 = (1, 0)$	where $e_1 = (1, 0)$ and $e_2 = (0, 1)$. Therefore, the set of all
rph8-g15q_F00075	5	■ 1.000	$e_2 = (0, 1)$	$e_2 = (0, 1)$	where $e_1 = (1, 0)$ and $e_2 = (0, 1)$. Therefore, the set of all
rph8-g15q_F00076	5	■ 1.000	$2(4^2 - 1) = 30$	$2(4^2 - 1) = 30$	a real vector space of dimension $2(4^2 - 1) = 30$.
rph8-g15q_F00077	5	■ 1.000	$\mathcal{L}(x)$	$\mathcal{L}(x)$	at the level of the generating elements $\mathcal{L}(x)$ because each V_g
rph8-g15q_F00078	5	■ 1.000	V_g	V_g	at the level of the generating elements $\mathcal{L}(x)$ because each V_g
rph8-g15q_F00079	5	■ 1.000	$g = (1, 0)$	$g = (1, 0)$	specific group elements. For $g = (1, 0)$, equivariance implies
rph8-g15q_F00080	5	■ 0.983	$\text{SWAP}_{L_1} \text{SWAP} = L_2$	$\text{SWAP}_{L_1} \text{SWAP} = L_2$	$\text{SWAP } L_1 \text{ SWAP} = L_2$ (and also $\text{SWAP } L_2 \text{ SWAP} = L_1$, but this
rph8-g15q_F00081	5	■ 0.983	$\text{SWAP}_{L_2} \text{SWAP} = L_1$	$\text{SWAP}_{L_2} \text{SWAP} = L_1$	$\text{SWAP } L_1 \text{ SWAP} = L_2$ (and also $\text{SWAP } L_2 \text{ SWAP} = L_1$, but this
rph8-g15q_F00082	5	■ 1.000	$g = (0, 1)$	$g = (0, 1)$	is equivalent to the first equation). For $g = (0, 1)$, equivari-
rph8-g15q_F00083	5	■ 1.000	$(X \otimes X)L_1(X \otimes X) = -L_1$	$(X \otimes X)L_1(X \otimes X) = -L_1$	ance implies $(X \otimes X)L_1(X \otimes X) = -L_1$ and $(X \otimes X)L_2(X \otimes$
rph8-g15q_F00084	5	■ 1.000	$(X \otimes X)L_2(X \otimes$	$(X \otimes X)L_2(X \otimes$	ance implies $(X \otimes X)L_1(X \otimes X) = -L_1$ and $(X \otimes X)L_2(X \otimes$
rph8-g15q_F00085	5	■ 1.000	$X) = -L_2$	$X) = -L_2$	$X) = -L_2$. However, the second of these follows from the

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rph8-g15q_F00086	5	0.538	$\left(L_{\{1\}}, L_{\{2\}}\right)$	(L_1, L_2)	tian matrices (L_1, L_2) to an 8-dimensional subspace of pairs
rph8-g15q_F00087	5	0.912	$L_{\{1\}}$	L_1	$(L_1, \text{SWAP } L_1 \text{ SWAP})$ due to the constraint (18). A basis for this
rph8-g15q_F00088	6	1.000	$\tau_{\{\mathcal{X}\}}$	$\tau_{\mathcal{X}}$	is a set $\tau_{\mathcal{X}}$ whose elements are certain subsets of $\mathcal{X}$, called
rph8-g15q_F00089	6	0.999	$\left(\mathcal{X}, \tau_{\{\mathcal{X}\}}\right)$	$(\mathcal{X}, \tau_{\mathcal{X}})$	set $\mathcal{X}$ is equipped with a topology $\tau_{\mathcal{X}}$, the pair $(\mathcal{X}, \tau_{\mathcal{X}})$ is
rph8-g15q_F00090	6	1.000	$\mathcal{A}$	$\mathcal{A}$	mal smooth atlas $\mathcal{A}$ on $\mathcal{X}$, and the triple $(\mathcal{X}, \tau_{\mathcal{X}}, \mathcal{A})$ is called a
rph8-g15q_F00091	6	1.000	$\left(\mathcal{X}, \tau_{\{\mathcal{X}\}}, \mathcal{A}\right)$	$(\mathcal{X}, \tau_{\mathcal{X}}, \mathcal{A})$	mal smooth atlas $\mathcal{A}$ on $\mathcal{X}$, and the triple $(\mathcal{X}, \tau_{\mathcal{X}}, \mathcal{A})$ is called a
rph8-g15q_F00092	6	0.755	$\mathcal{H}=\mathbb{C}^d$	$\mathcal{H} = \mathbb{C}^d$	$\mathcal{H} = \mathbb{C}^d$ is the complex projective space $\mathbb{CP}^{d-1}$, which also
rph8-g15q_F00093	6	0.755	$\mathbb{C}^d$	$\mathbb{CP}^{d-1}$	$\mathcal{H} = \mathbb{C}^d$ is the complex projective space $\mathbb{CP}^{d-1}$, which also
rph8-g15q_F00094	6	1.000	$\mathcal{X}=\mathbb{R}^d$	$\mathcal{X} = \mathbb{R}^d$	for Hamiltonian evolution. In the case that $\mathcal{X} = \mathbb{R}^d$ is a Eu-
rph8-g15q_F00095	6	1.000	$x=\left(x_{\{1\}}, \ldots, x_{\{d\}}\right) \in$	$x = (x_1, \dots, x_d) \in$	clidean feature space, each component of $x = (x_1, \dots, x_d) \in$
rph8-g15q_F00096	6	1.000	$X_{\{k\}}$	X_k	where X_k is the Pauli X gate acting on the k th qubit. The
rph8-g15q_F00097	6	1.000	k	k	where X_k is the Pauli X gate acting on the k th qubit. The
rph8-g15q_F00098	6	0.975	$\rho: \mathbb{R}^d \rightarrow \mathcal{S}(\mathbb{C}^{2^d})$	$\rho : \mathbb{R}^d \rightarrow \mathcal{S}(\mathbb{C}^{2^d})$	associated quantum state encoding $\rho : \mathbb{R}^d \rightarrow \mathcal{S}(\mathbb{C}^{2^d})$ is then
rph8-g15q_F00099	6	1.000	Y	Y	Note that the Pauli X gate may be replaced with a Pauli Y or
rph8-g15q_F00100	6	1.000	$\vec{n} \cdot \vec{\sigma}$	$\vec{n} \cdot \vec{\sigma}$	$\vec{n} \cdot \vec{\sigma}$, where $\vec{n}$ is some unit vector in $\mathbb{R}^3$ and $\vec{\sigma} = (X, Y, Z)$
rph8-g15q_F00101	6	1.000	$\vec{n}$	$\vec{n}$	$\vec{n} \cdot \vec{\sigma}$, where $\vec{n}$ is some unit vector in $\mathbb{R}^3$ and $\vec{\sigma} = (X, Y, Z)$
rph8-g15q_F00102	6	1.000	$\mathbb{R}^3$	$\mathbb{R}^3$	$\vec{n} \cdot \vec{\sigma}$, where $\vec{n}$ is some unit vector in $\mathbb{R}^3$ and $\vec{\sigma} = (X, Y, Z)$
rph8-g15q_F00103	6	1.000	$\vec{\sigma}=(X, Y, Z)$	$\vec{\sigma} = (X, Y, Z)$	$\vec{n} \cdot \vec{\sigma}$, where $\vec{n}$ is some unit vector in $\mathbb{R}^3$ and $\vec{\sigma} = (X, Y, Z)$
rph8-g15q_F00104	6	1.000	d	d	tum encodings map a feature space of d dimensions into d
rph8-g15q_F00105	7	1.000	$x \in \mathbb{R}^{2^d} \setminus \{0\}$	$x \in \mathbb{R}^{2^d} \setminus \{0\}$	[59]. By definition, it takes a vector $x \in \mathbb{R}^{2^d} \setminus \{0\}$ to a vector
rph8-g15q_F00106	7	0.999	$\chi(x)$	$\chi(x)$	$\chi(x)$ in $\mathbb{C}^{2^d}$, the Hilbert space of d qubits, and then normalizes
rph8-g15q_F00107	7	0.999	$\mathbb{C}^{2^d}$	$\mathbb{C}^{2^d}$	$\chi(x)$ in $\mathbb{C}^{2^d}$, the Hilbert space of d qubits, and then normalizes
rph8-g15q_F00108	7	1.000	$ \Phi(x)\rangle$	$ \Phi(x)\rangle$	it to obtain a genuine quantum state $ \Phi(x)\rangle$. To express this
rph8-g15q_F00109	7	1.000	$x=(x_{\{0\}}, x_{\{1\}}, \ldots, x_{\{2^d-1\}}) \in \mathbb{R}^{2^d}$	$x = (x_0, x_1, \dots, x_{2^d-1}) \in \mathbb{R}^{2^d}$	as a succinct formula, if $x = (x_0, x_1, \dots, x_{2^d-1}) \in \mathbb{R}^{2^d}$, then
rph8-g15q_F00110	7	0.998	n	n	sequence n of d 0s and 1s. For example, if $d = 2$, then n
rph8-g15q_F00111	7	0.998	$d0 \text{ s}$	$d0 \text{ s}$	sequence n of d 0s and 1s. For example, if $d = 2$, then n
rph8-g15q_F00112	7	0.998	$d=2$	$d = 2$	sequence n of d 0s and 1s. For example, if $d = 2$, then n
rph8-g15q_F00113	7	0.999	$n \in \{00, 01, 10, 11\}$	$n \in \{00, 01, 10, 11\}$	has four possibilities: $n \in \{00, 01, 10, 11\}$. Then, amplitude
rph8-g15q_F00114	7	1.000	$\ \chi(x)\ $	$\ \chi(x)\ $	where $\ \chi(x)\ $ denotes the Euclidean norm of $\chi(x)$. Since the
rph8-g15q_F00115	7	1.000	S^{2^d-1}	S^{2^d-1}	the data first onto the sphere S^{2^d-1} inside $\mathbb{R}^{2^d}$ and then map
rph8-g15q_F00116	7	1.000	$\mathbb{R}^{2^d}$	$\mathbb{R}^{2^d}$	the data first onto the sphere S^{2^d-1} inside $\mathbb{R}^{2^d}$ and then map
rph8-g15q_F00117	7	0.999	$\log_2(2^d)=d$	$\log_2(2^d) = d$	the form $\mathbb{R}^{2^d}$ is encoded into a Hilbert space of $\log_2(2^d) = d$
rph8-g15q_F00118	7	1.000	S^{2^d}	S^{2^d}	the original data and project onto the unit sphere S^{2^d} (instead
rph8-g15q_F00119	7	1.000	$\frac{\pi}{4}$	$\frac{\pi}{4}$	hemisphere of S^{2^d} within a polar angle of $\frac{\pi}{4}$, while the points
rph8-g15q_F00120	7	0.999	$\frac{\pi}{2}$	$\frac{\pi}{2}$	hemisphere of S^{2^d} with a polar angle between $\frac{\pi}{4}$ and $\frac{\pi}{2}$, thus
rph8-g15q_F00121	7	0.952	$x_{\{0\}}$	x_0	FIG. 5. Although data x_0 in the interval $[-1, 1]$ get mapped into
rph8-g15q_F00122	7	1.000	S^1	S^1	the northern hemisphere of S^1 within polar angle $\frac{\pi}{4}$, all other data
rph8-g15q_F00123	7	1.000	$[-6, 6] \cap \mathbb{Z}$	$[-6, 6] \cap \mathbb{Z}$	$[-6, 6] \cap \mathbb{Z}$.
rph8-g15q_F00124	8	0.978	$\mathbb{R}$	$\mathbb{R}$	FIG. 6. Data x_0 in $\mathbb{R}$ get mapped into the probability 1-simplex.
rph8-g15q_F00125	8	0.706	$(\mathcal{X}, d_{\mathcal{X}})$	$(\mathcal{X}, d_{\mathcal{X}})$	<i>Definition 5.</i> A <i>metric space</i> is a pair $(\mathcal{X}, d_{\mathcal{X}})$, where $\mathcal{X}$
rph8-g15q_F00126	8	1.000	$d_{\mathcal{X}}: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$	$d_{\mathcal{X}} : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$	is a set and $d_{\mathcal{X}} : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a function satisfying the
rph8-g15q_F00127	8	1.000	$d_{\mathcal{X}}(x_1, x_2)=0$	$d_{\mathcal{X}}(x_1, x_2) = 0$	(1) $d_{\mathcal{X}}(x_1, x_2) = 0$ if and only if $x_1 = x_2$.
rph8-g15q_F00128	8	1.000	$x_1=x_2$	$x_1 = x_2$	(1) $d_{\mathcal{X}}(x_1, x_2) = 0$ if and only if $x_1 = x_2$.
rph8-g15q_F00129	8	1.000	$d_{\mathcal{X}}(x_1, x_2)=d_{\mathcal{X}}(x_2, x_1)$	$d_{\mathcal{X}}(x_1, x_2) = d_{\mathcal{X}}(x_2, x_1)$	(2) $d_{\mathcal{X}}(x_1, x_2) = d_{\mathcal{X}}(x_2, x_1)$ for all $x_1, x_2 \in \mathcal{X}$.
rph8-g15q_F00130	8	1.000	$x_1, x_2 \in \mathcal{X}$	$x_1, x_2 \in \mathcal{X}$	(2) $d_{\mathcal{X}}(x_1, x_2) = d_{\mathcal{X}}(x_2, x_1)$ for all $x_1, x_2 \in \mathcal{X}$.

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rph8-g15q_F00131	8	■ 1.000	$d_{\mathcal{X}}(\mathbf{x}_1, \mathbf{x}_2) \leq d_{\mathcal{X}}(\mathbf{x}_1, \mathbf{x}_3) + d_{\mathcal{X}}(\mathbf{x}_2, \mathbf{x}_3)$	$d_{\mathcal{X}}(x_1, x_2) \leq d_{\mathcal{X}}(x_1, x_3) + d_{\mathcal{X}}(x_2, x_3)$	(3) $d_{\mathcal{X}}(x_1, x_2) \leq d_{\mathcal{X}}(x_1, x_3) + d_{\mathcal{X}}(x_2, x_3)$ for all x_1, x_2, x_3
rph8-g15q_F00132	8	■ 1.000	$\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$	x_1, x_2, x_3	(3) $d_{\mathcal{X}}(x_1, x_2) \leq d_{\mathcal{X}}(x_1, x_3) + d_{\mathcal{X}}(x_2, x_3)$ for all x_1, x_2, x_3
rph8-g15q_F00133	8	■ 1.000	$\mathbf{x} \in \mathcal{X}$	$\in \mathcal{X}$	$\in \mathcal{X}$ (the <i>triangle inequality</i>).
rph8-g15q_F00134	8	■ 1.000	$d_{\mathcal{X}}$	$d_{\mathcal{X}}$	Such a function $d_{\mathcal{X}}$ is called a <i>distance function</i> , or <i>metric</i> ,
rph8-g15q_F00135	8	■ 1.000	$X \subseteq \mathcal{X}$	$X \subseteq \mathcal{X}$	(1) If $X \subseteq \mathcal{X}$ is a finite subset of $\mathcal{X}$, which is to be
rph8-g15q_F00136	8	■ 0.969	(X, d_X)	(X, d_X)	distances (X, d_X) to first construct a combinatorial object $K_{\bullet}(X, d_X)$,
rph8-g15q_F00137	8	■ 0.969	$K_{\bullet}(X, d_X)$	$K_{\bullet}(X, d_X)$	distances (X, d_X) to first construct a combinatorial object $K_{\bullet}(X, d_X)$,
rph8-g15q_F00138	8	■ 1.000	$H_{\bullet}(X, d_X)$	$H_{\bullet}(X, d_X)$	which is then used to construct an algebraic object $H_{\bullet}(X, d_X)$, which
rph8-g15q_F00139	8	■ 1.000	d_X	d_X	metric $d_{\mathcal{X}}$ on it, then X acquires a metric d_X by the formula
rph8-g15q_F00140	8	■ 1.000	$d_X(\mathbf{x}_1, \mathbf{x}_2) = d_{\mathcal{X}}(\mathbf{x}_1, \mathbf{x}_2)$	$d_X(x_1, x_2) = d_{\mathcal{X}}(x_1, x_2)$	$d_X(x_1, x_2) = d_{\mathcal{X}}(x_1, x_2)$ for all $x_1, x_2 \in X$.
rph8-g15q_F00141	8	■ 1.000	$\mathbf{x}_1, \mathbf{x}_2 \in X$	$x_1, x_2 \in X$	$d_X(x_1, x_2) = d_{\mathcal{X}}(x_1, x_2)$ for all $x_1, x_2 \in X$.
rph8-g15q_F00142	8	■ 0.953	X, d_X	X, d_X	(2) From (X, d_X) , one can construct a combinatorial ob-
rph8-g15q_F00143	8	■ 0.829	$H_{\bullet}(X, d_X)$ [45,86,87]	$H_{\bullet}(X, d_X)$ [45,86,87]	spaces and homologies $H_{\bullet}(X, d_X)$ [45,86,87].
rph8-g15q_F00144	8	■ 0.920	$\text{dgm}_{\bullet}(X, d_X)$	$\text{dgm}_{\bullet}(X, d_X)$	numerical quantities, the persistence diagrams $\text{dgm}_{\bullet}(X, d_X)$
rph8-g15q_F00145	8	■ 1.000	S	S	rather than the raw data. Briefly, if S denotes a simplicial
rph8-g15q_F00146	8	■ 1.000	ϵ	ϵ	distance threshold ϵ , then each k -simplex $s \in S$ is uniquely
rph8-g15q_F00147	8	■ 1.000	$s \in S$	$s \in S$	distance threshold ϵ , then each k -simplex $s \in S$ is uniquely
rph8-g15q_F00148	8	■ 1.000	$k+1$	$k+1$	determined by its $k+1$ vertices. Supposing that there is a
rph8-g15q_F00149	8	■ 1.000	$v_1, v_2, \dots, v_n$	$v_1, v_2, \dots, v_n$	total ordering on the vertices, $v_1, v_2, \dots, v_n$, such a simplex
rph8-g15q_F00150	8	■ 1.000	s	s	s can therefore be expressed as $s = \{v_{i_1}, \dots, v_{i_{k+1}}\}$, where
rph8-g15q_F00151	8	■ 1.000	$s = \{v_{i_1}, \dots, v_{i_{k+1}}\}$	$s = \{v_{i_1}, \dots, v_{i_{k+1}}\}$	s can therefore be expressed as $s = \{v_{i_1}, \dots, v_{i_{k+1}}\}$, where
rph8-g15q_F00152	8	■ 0.863	$i_1 < \dots < i_{k+1}$	$i_1 < \dots < i_{k+1}$	$i_1 < \dots < i_{k+1}$ and the distinct vertices $v_{i_1}, \dots, v_{i_{k+1}}$ define
rph8-g15q_F00153	8	■ 0.863	$v_{i_1}, \dots, v_{i_{k+1}}$	$v_{i_1}, \dots, v_{i_{k+1}}$	$i_1 < \dots < i_{k+1}$ and the distinct vertices $v_{i_1}, \dots, v_{i_{k+1}}$ define
rph8-g15q_F00154	8	■ 1.000	$ s\rangle$	$ s\rangle$	the simplex s . Therefore, the quantum state $ s\rangle$ associated with
rph8-g15q_F00155	8	■ 1.000	$i_1, \dots, i_{k+1}$	$i_1, \dots, i_{k+1}$	the simplex s would be the n -qubit state with the $i_1, \dots, i_{k+1}$
rph8-g15q_F00156	8	■ 1.000	$ 1\rangle$	$ 1\rangle$	qubits in state $ 1\rangle$ and all other qubits in state $ 0\rangle$ (this is
rph8-g15q_F00157	8	■ 1.000	$ 0\rangle$	$ 0\rangle$	qubits in state $ 1\rangle$ and all other qubits in state $ 0\rangle$ (this is
rph8-g15q_F00158	8	■ 1.000	$n=7$	$n=7$	tail in Appendix E). For example, for $n=7$ and $k=3$, the
rph8-g15q_F00159	8	■ 1.000	$k=3$	$k=3$	tail in Appendix E). For example, for $n=7$ and $k=3$, the
rph8-g15q_F00160	8	■ 1.000	$s = \{v_2, v_4, v_5\}$	$s = \{v_2, v_4, v_5\}$	2-simplex $s = \{v_2, v_4, v_5\}$ would be represented by the state
rph8-g15q_F00161	8	■ 1.000	$ s\rangle = 0101100\rangle$	$ s\rangle = 0101100\rangle$	$ s\rangle = 0101100\rangle$.
rph8-g15q_F00162	8	■ 1.000	f	f	<i>Definition 6.</i> A <i>distance nonincreasing</i> function f :
rph8-g15q_F00163	8	■ 1.000	$(\mathcal{X}, d_{\mathcal{X}}) \rightarrow (\mathcal{Y}, d_{\mathcal{Y}})$	$(\mathcal{X}, d_{\mathcal{X}}) \rightarrow (\mathcal{Y}, d_{\mathcal{Y}})$	$(\mathcal{X}, d_{\mathcal{X}}) \rightarrow (\mathcal{Y}, d_{\mathcal{Y}})$ from one metric space $(\mathcal{X}, d_{\mathcal{X}})$ to
rph8-g15q_F00164	8	■ 1.000	$(\mathcal{Y}, d_{\mathcal{Y}})$	$(\mathcal{Y}, d_{\mathcal{Y}})$	another $(\mathcal{Y}, d_{\mathcal{Y}})$ is a function $f: \mathcal{X} \rightarrow \mathcal{Y}$ that satisfies
rph8-g15q_F00165	8	■ 1.000	$f: \mathcal{X} \rightarrow \mathcal{Y}$	$f: \mathcal{X} \rightarrow \mathcal{Y}$	another $(\mathcal{Y}, d_{\mathcal{Y}})$ is a function $f: \mathcal{X} \rightarrow \mathcal{Y}$ that satisfies
rph8-g15q_F00166	8	■ 0.780	$d_{\mathcal{Y}}(f(\mathbf{x}_1), f(\mathbf{x}_2)) \leq d_{\mathcal{X}}(\mathbf{x}_1, \mathbf{x}_2)$	$d_{\mathcal{Y}}(f(x_1), f(x_2)) \leq d_{\mathcal{X}}(x_1, x_2)$	$d_{\mathcal{Y}}(f(x_1), f(x_2)) \leq d_{\mathcal{X}}(x_1, x_2)$ for all $x_1, x_2 \in \mathcal{X}$. An <i>embed-</i>
rph8-g15q_F00167	9	■ 1.000	$d_{\mathcal{Y}}(f(\mathbf{x}_1), f(\mathbf{x}_2)) = d_{\mathcal{X}}(\mathbf{x}_1, \mathbf{x}_2)$	$d_{\mathcal{Y}}(f(x_1), f(x_2)) = d_{\mathcal{X}}(x_1, x_2)$	satisfies $d_{\mathcal{Y}}(f(x_1), f(x_2)) = d_{\mathcal{X}}(x_1, x_2)$ for all $x_1, x_2 \in \mathcal{X}$; i.e.,
rph8-g15q_F00168	9	■ 1.000	$f: (X, d_X) \rightarrow (\mathcal{Y}, d_{\mathcal{Y}})$	$f: (X, d_X) \rightarrow (\mathcal{Y}, d_{\mathcal{Y}})$	metric d_X from $d_{\mathcal{X}}$. If $f: (X, d_X) \rightarrow (\mathcal{Y}, d_{\mathcal{Y}})$ is an embed-
rph8-g15q_F00169	9	■ 1.000	$Y := f(X)$	$Y := f(X)$	ding, let $Y := f(X)$ be the image of X under f equipped
rph8-g15q_F00170	9	■ 1.000	d_Y	d_Y	with the induced metric d_Y from $d_{\mathcal{Y}}$. Then, the filtered sim-
rph8-g15q_F00171	9	■ 1.000	d_Y	d_Y	with the induced metric d_Y from $d_{\mathcal{Y}}$. Then, the filtered sim-
rph8-g15q_F00172	9	■ 1.000	$K_{\bullet}(Y, d_Y)$	$K_{\bullet}(Y, d_Y)$	plicial complexes $K_{\bullet}(X, d_X)$ and $K_{\bullet}(Y, d_Y)$ are isomorphic,
rph8-g15q_F00173	9	■ 1.000	$H_{\bullet}(Y, d_Y)$	$H_{\bullet}(Y, d_Y)$	the persistence vector spaces $H_{\bullet}(X, d_X)$ and $H_{\bullet}(Y, d_Y)$ are
rph8-g15q_F00174	9	■ 1.000	$\text{dgm}_{\bullet}(Y, d_Y)$	$\text{dgm}_{\bullet}(Y, d_Y)$	and $\text{dgm}_{\bullet}(Y, d_Y)$ are all isomorphic.
rph8-g15q_F00175	9	■ 0.988	$\mathcal{X} \xrightarrow{f} \mathcal{Y}$	$\mathcal{X} \xrightarrow{f} \mathcal{Y}$	<i>Lemma 1.</i> Let $\mathcal{X}$ be a set, $(\mathcal{Y}, d_{\mathcal{Y}})$ a metric space, $\mathcal{X} \xrightarrow{f} \mathcal{Y}$

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rph8-g15q_F00176	9	■ 1.000	$d_{\mathcal{X}}: \mathcal{X} \times \mathcal{X} \rightarrow [0, \infty)$	$d_{\mathcal{X}}: \mathcal{X} \times \mathcal{X} \rightarrow [0, \infty)$	a function, and $d_{\mathcal{X}}: \mathcal{X} \times \mathcal{X} \rightarrow [0, \infty)$ the function
rph8-g15q_F00177	9	■ 1.000	$\left(\mathcal{X}, d_{\mathcal{X}}\right) \xrightarrow{f} \left(\mathcal{Y}, d_{\mathcal{Y}}\right)$	$(\mathcal{X}, d_{\mathcal{X}}) \xrightarrow{f} (\mathcal{Y}, d_{\mathcal{Y}})$	only if f is one to one. In such a case, $(\mathcal{X}, d_{\mathcal{X}}) \xrightarrow{f} (\mathcal{Y}, d_{\mathcal{Y}})$ is
rph8-g15q_F00178	9	■ 1.000	L_p	L_p	metrics induced by the L_p norm [91]) and (2) a family of
rph8-g15q_F00179	10	■ 0.999	$\left(\mathcal{X}, d_{\mathcal{X}_{\text{true}}}\right)$	$(\mathcal{X}, d_{\mathcal{X}_{\text{true}}})$	such that $(\mathcal{X}, d_{\mathcal{X}})$ is close to $(\mathcal{X}, d_{\mathcal{X}_{\text{true}}})$, where $d_{\mathcal{X}_{\text{true}}}$ is some
rph8-g15q_F00180	10	■ 0.999	$d_{\mathcal{X}_{\text{true}}}$	$d_{\mathcal{X}_{\text{true}}}$	such that $(\mathcal{X}, d_{\mathcal{X}})$ is close to $(\mathcal{X}, d_{\mathcal{X}_{\text{true}}})$, where $d_{\mathcal{X}_{\text{true}}}$ is some
rph8-g15q_F00181	10	■ 1.000	σ	σ	<i>Definition 8.</i> Fix a Hilbert space $\mathcal{H}$ and let ρ and σ be two
rph8-g15q_F00182	10	■ 0.518	ℓ_1	ℓ_1	density matrices on $\mathcal{H}$. The <i>trace/ℓ_1 distance</i> between ρ and
rph8-g15q_F00183	10	■ 1.000	ℓ_2	ℓ_2	The <i>Hilbert-Schmidt/Frobenius/ℓ_2 distance</i> is
rph8-g15q_F00184	10	■ 1.000	$\rho, \sigma \in \mathcal{S}(\mathcal{H})$	$\rho, \sigma \in \mathcal{S}(\mathcal{H})$	denotes the <i>fidelity</i> between the states $\rho, \sigma \in \mathcal{S}(\mathcal{H})$.
rph8-g15q_F00185	10	■ 1.000	$d_{\mathcal{H}}$	$d_{\mathcal{H}}$	references therein. Let $d_{\mathcal{H}}$ be the notation for any one of
rph8-g15q_F00186	10	■ 1.000	Θ	Θ	One may additionally add on a parameter space Θ to define
rph8-g15q_F00187	10	■ 1.000	$\rho: \mathcal{X} \times \Theta \rightarrow \mathcal{S}(\mathcal{H})$	$\rho: \mathcal{X} \times \Theta \rightarrow \mathcal{S}(\mathcal{H})$	a function of the form $\rho: \mathcal{X} \times \Theta \rightarrow \mathcal{S}(\mathcal{H})$, from which a
rph8-g15q_F00188	10	■ 1.000	$\theta \in \Theta$	$\theta \in \Theta$	specific parameter $\theta \in \Theta$ is chosen to optimize some machine
rph8-g15q_F00189	10	■ 1.000	$\mathcal{X}'$	$\mathcal{X}'$	for mathematical objects $\mathcal{X}'$ and $\mathcal{S}(\mathcal{H})'$ in a particular <i>cat-</i>
rph8-g15q_F00190	10	■ 1.000	$\mathcal{S}(\mathcal{H})'$	$\mathcal{S}(\mathcal{H})'$	for mathematical objects $\mathcal{X}'$ and $\mathcal{S}(\mathcal{H})'$ in a particular <i>cat-</i>
rph8-g15q_F00191	10	■ 1.000	$\rho': \mathcal{X}' \rightarrow \mathcal{S}(\mathcal{H})'$	$\rho': \mathcal{X}' \rightarrow \mathcal{S}(\mathcal{H})'$	any structure, corresponds to a <i>morphism</i> $\rho': \mathcal{X}' \rightarrow \mathcal{S}(\mathcal{H})'$
rph8-g15q_F00192	10	■ 1.000	ρ'	ρ'	in that category. This mapping ρ' can be thought of as an
rph8-g15q_F00193	11	■ 1.000	$\mathcal{X} = [-2, 2]$	$\mathcal{X} = [-2, 2]$	FIG. 8. (a) Two classes of data on $\mathcal{X} = [-2, 2]$ are presented. One class has orange data points labeled ○ clustered near 0 and the other class
rph8-g15q_F00194	11	■ 0.974	$\rho: [-2, 2] \rightarrow \mathcal{S}(\mathbb{C}^2)$	$\rho: [-2, 2] \rightarrow \mathcal{S}(\mathbb{C}^2)$	has blue data points labeled ✕ clustered near 2 and -2 . (b) The image of the quantum encoding $\rho: [-2, 2] \rightarrow \mathcal{S}(\mathbb{C}^2)$, with $\mathcal{S}(\mathbb{C}^2)$ visualized
rph8-g15q_F00195	11	■ 0.974	$\mathcal{S}(\mathbb{C}^2)$	$\mathcal{S}(\mathbb{C}^2)$	has blue data points labeled ✕ clustered near 2 and -2 . (b) The image of the quantum encoding $\rho: [-2, 2] \rightarrow \mathcal{S}(\mathbb{C}^2)$, with $\mathcal{S}(\mathbb{C}^2)$ visualized
rph8-g15q_F00196	11	■ 0.614	$\theta_1 = 0.31, \theta_2 = 1.48, \theta_3 = 0$	$\theta_1 = 0.31, \theta_2 = 1.48, \theta_3 = 0$	as the Bloch ball and the encoding ρ defined in Eq. (D10) using the optimized parameters $\theta_1 = 0.31, \theta_2 = 1.48, \theta_3 = 0$, and $\theta_4 = 0$. (c) A
rph8-g15q_F00197	11	■ 0.614	$\theta_4 = 0$	$\theta_4 = 0$	as the Bloch ball and the encoding ρ defined in Eq. (D10) using the optimized parameters $\theta_1 = 0.31, \theta_2 = 1.48, \theta_3 = 0$, and $\theta_4 = 0$. (c) A
rph8-g15q_F00198	11	■ 0.993	$\rho: \mathcal{X} \rightarrow \mathcal{S}(\mathbb{C}^2)$	$\rho: \mathcal{X} \rightarrow \mathcal{S}(\mathbb{C}^2)$	visualization of the pullback metric from the quantum encoding $\rho: \mathcal{X} \rightarrow \mathcal{S}(\mathbb{C}^2)$ back onto the domain $\mathcal{X}$. The metric has been uniformly
rph8-g15q_F00199	11	■ 0.679	$f:$	$f:$	that for any two sets $\mathcal{X}$ and $\mathcal{Y}$, there is a set of functions $f:$
rph8-g15q_F00200	11	■ 1.000	$\mathcal{X} \rightarrow \mathcal{Y}$	$\mathcal{X} \rightarrow \mathcal{Y}$	$\mathcal{X} \rightarrow \mathcal{Y}$ from $\mathcal{X}$ to $\mathcal{Y}$. Functions, though quite minimalistic,
rph8-g15q_F00201	11	■ 1.000	$g: \mathcal{Y} \rightarrow \mathcal{Z}$	$g: \mathcal{Y} \rightarrow \mathcal{Z}$	Namely, for any two functions $f: \mathcal{X} \rightarrow \mathcal{Y}$ and $g: \mathcal{Y} \rightarrow \mathcal{Z}$,
rph8-g15q_F00202	11	■ 1.000	$g \circ f: \mathcal{X} \rightarrow \mathcal{Z}$	$g \circ f: \mathcal{X} \rightarrow \mathcal{Z}$	their composite is also a function $g \circ f: \mathcal{X} \rightarrow \mathcal{Z}$. Moreover,
rph8-g15q_F00203	11	■ 1.000	$f: \mathcal{X} \rightarrow \mathcal{Y}, g: \mathcal{Y} \rightarrow \mathcal{Z}$	$f: \mathcal{X} \rightarrow \mathcal{Y}, g: \mathcal{Y} \rightarrow \mathcal{Z}$	functions $f: \mathcal{X} \rightarrow \mathcal{Y}, g: \mathcal{Y} \rightarrow \mathcal{Z}$, and $h: \mathcal{Z} \rightarrow \mathcal{W}$, the com-
rph8-g15q_F00204	11	■ 1.000	$h: \mathcal{Z} \rightarrow \mathcal{W}$	$h: \mathcal{Z} \rightarrow \mathcal{W}$	functions $f: \mathcal{X} \rightarrow \mathcal{Y}, g: \mathcal{Y} \rightarrow \mathcal{Z}$, and $h: \mathcal{Z} \rightarrow \mathcal{W}$, the com-
rph8-g15q_F00205	11	■ 1.000	$h \circ (g \circ f)$	$h \circ (g \circ f)$	posites $h \circ (g \circ f)$ and $(h \circ g) \circ f$ are equal. Finally, for each
rph8-g15q_F00206	11	■ 1.000	$(h \circ g) \circ f$	$(h \circ g) \circ f$	posites $h \circ (g \circ f)$ and $(h \circ g) \circ f$ are equal. Finally, for each
rph8-g15q_F00207	11	■ 1.000	$\text{id}_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X}$	$\text{id}_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X}$	set $\mathcal{X}$, there is a particular function $\text{id}_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X}$ defined by
rph8-g15q_F00208	11	■ 0.996	$\text{id}_{\mathcal{X}}(x) = x$	$\text{id}_{\mathcal{X}}(x) = x$	$\text{id}_{\mathcal{X}}(x) = x$ for all $x \in \mathcal{X}$ that acts as a unit/identity for this
rph8-g15q_F00209	11	■ 0.612	$\text{id}_{\mathcal{Y}} \circ f = f \circ \text{id}_{\mathcal{X}} = f$	$\text{id}_{\mathcal{Y}} \circ f = f \circ \text{id}_{\mathcal{X}} = f$	composition since $\text{id}_{\mathcal{Y}} \circ f = f \circ \text{id}_{\mathcal{X}} = f$ for all functions
rph8-g15q_F00210	11	■ 1.000	$\mathbf{C}$	$\mathbf{C}$	<i>Definition 10.</i> A category $\mathbf{C}$ consists of
rph8-g15q_F00211	11	■ 0.996	$\mathcal{X}, \mathcal{Y}, \dots$	$\mathcal{X}, \mathcal{Y}, \dots$	(1) a collection of <i>objects</i> , denoted as $\mathcal{X}, \mathcal{Y}, \dots$
rph8-g15q_F00212	11	■ 1.000	$\mathcal{X}, \mathcal{Y}$	$\mathcal{X}, \mathcal{Y}$	(2) for every pair of objects $\mathcal{X}, \mathcal{Y}$, a set of <i>morphisms</i> from
rph8-g15q_F00213	11	■ 0.957	$f: \mathcal{X} \rightarrow \mathcal{Y}, g: \mathcal{Y} \rightarrow \mathcal{Z}$	$f: \mathcal{X} \rightarrow \mathcal{Y}, g: \mathcal{Y} \rightarrow \mathcal{Z}$,	composable morphisms $f: \mathcal{X} \rightarrow \mathcal{Y}, g: \mathcal{Y} \rightarrow \mathcal{Z}$, and
rph8-g15q_F00214	12	■ 0.737	$f \circ \text{id}_{\mathcal{X}} =$	$f \circ \text{id}_{\mathcal{X}} =$	$\mathcal{X}$ there exists a morphism $\text{id}_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X}$ such that $f \circ \text{id}_{\mathcal{X}} =$
rph8-g15q_F00215	12	■ 0.994	$\text{id}_{\mathcal{Y}} \circ f = f$	$\text{id}_{\mathcal{Y}} \circ f = f$	$\text{id}_{\mathcal{Y}} \circ f = f$ for all morphisms $f: \mathcal{X} \rightarrow \mathcal{Y}$.
rph8-g15q_F00216	12	■ 0.932	$\text{id}_{\mathcal{X}}$	$\text{id}_{\mathcal{X}}$	(3) For each G -set $(\mathcal{X}, \alpha)$, the identity function $\text{id}_{\mathcal{X}}$ on $\mathcal{X}$
rph8-g15q_F00217	12	■ 0.985	emb	emb	category $\mathbf{Met}^{\text{emb}}$ whose objects are metric spaces, where a
rph8-g15q_F00218	12	■ 0.958	nif	nif	<i>functions</i>). There is a category $\mathbf{Met}^{\text{nif}}$ whose objects are met-

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rph8-g15q_F00219	12	0.738	$\mathbf{Met}^{\mathrm{nif}}$	$\mathbf{Met}^{\mathrm{nif}}$	ric spaces $(\mathcal{X}, d_{\mathcal{X}})$. A morphism in $\mathbf{Met}^{\mathrm{nif}}$ between metric
rph8-g15q_F00220	12	0.982	$\{\}^{\mathrm{inf}}$	inf	<i>subcategory</i> of $\mathbf{Met}^{\mathrm{inf}}$, meaning that all objects and mor-
rph8-g15q_F00221	12	0.876	$\{\}^{\mathrm{nif}}$	nif	phisms of $\mathbf{Met}^{\mathrm{emb}}$ are also objects and morphisms of $\mathbf{Met}^{\mathrm{nif}}$,
rph8-g15q_F00222	12	1.000	$\rho: (\mathcal{X}, \alpha) \rightarrow (\mathcal{S}(\mathcal{H}), \mathrm{Ad}_V)$	$\rho: (\mathcal{X}, \alpha) \rightarrow (\mathcal{S}(\mathcal{H}), \mathrm{Ad}_V)$	a G -equivariant function $\rho: (\mathcal{X}, \alpha) \rightarrow (\mathcal{S}(\mathcal{H}), \mathrm{Ad}_V)$ which,
rph8-g15q_F00223	12	0.999	$F: G$	$F: G$	denote this assignment by $F: G\text{-}\mathbf{Set} \rightarrow \mathbf{Set}$, so that for an
rph8-g15q_F00224	13	1.000	$F(\mathcal{X}, \alpha) = \mathcal{X}$	$F(\mathcal{X}, \alpha) = \mathcal{X}$	arbitrary G -set $(\mathcal{X}, \alpha)$, we have $F(\mathcal{X}, \alpha) = \mathcal{X}$ and for any G -
rph8-g15q_F00225	13	1.000	$F(f) = f$	$F(f) = f$	equivariant function f , we have $F(f) = f$. Then, F behaves
rph8-g15q_F00226	13	1.000	F	F	equivariant function f , we have $F(f) = f$. Then, F behaves
rph8-g15q_F00227	13	1.000	g	g	and g , we have, somewhat trivially,
rph8-g15q_F00228	13	1.000	$x \mapsto x$	$x \mapsto x$	finied by $x \mapsto x$ for all $x \in \mathcal{X}$. So, forgetting the group action
rph8-g15q_F00229	13	1.000	$F: \mathbf{C} \rightarrow \mathbf{D}$	$F: \mathbf{C} \rightarrow \mathbf{D}$	<i>Definition 11.</i> A <i>functor</i> $F: \mathbf{C} \rightarrow \mathbf{D}$ between categories
rph8-g15q_F00230	13	0.582	$F(\mathcal{X})$	$F(\mathcal{X})$	(1) an object $F(\mathcal{X})$ in $\mathbf{D}$ for every object $\mathcal{X}$ in $\mathbf{C}$
rph8-g15q_F00231	13	0.582	$\mathbf{D}$	$\mathbf{D}$	(1) an object $F(\mathcal{X})$ in $\mathbf{D}$ for every object $\mathcal{X}$ in $\mathbf{C}$
rph8-g15q_F00232	13	0.988	$F(f): F(\mathcal{X}) \rightarrow F(\mathcal{Y})$	$F(f): F(\mathcal{X}) \rightarrow F(\mathcal{Y})$	(2) a morphism $F(f): F(\mathcal{X}) \rightarrow F(\mathcal{Y})$ in $\mathbf{D}$ for every
rph8-g15q_F00233	13	1.000	$F(g \circ f) =$	$F(g \circ f) =$	(1) Composition is preserved; that is, $F(g \circ f) =$
rph8-g15q_F00234	13	1.000	$F(g) \circ F(f)$	$F(g) \circ F(f)$	$F(g) \circ F(f)$ for all morphisms $f: \mathcal{X} \rightarrow \mathcal{Y}$ and $g: \mathcal{Y} \rightarrow \mathcal{Z}$
rph8-g15q_F00235	13	0.452	$F(\mathrm{id}_{\mathcal{X}}) = \mathrm{id}_{F(\mathcal{X})}$	$F(\mathrm{id}_{\mathcal{X}}) = \mathrm{id}_{F(\mathcal{X})}$	(2) Identities are preserved; that is, $F(\mathrm{id}_{\mathcal{X}}) = \mathrm{id}_{F(\mathcal{X})}$ for
rph8-g15q_F00236	13	1.000	$\mathbf{C}(\mathcal{X}, \mathcal{Y})$	$\mathbf{C}(\mathcal{X}, \mathcal{Y})$	properties that functors may satisfy. Let $\mathbf{C}(\mathcal{X}, \mathcal{Y})$ denote the
rph8-g15q_F00237	13	1.000	$\mathbf{C}(\mathcal{X}, \mathcal{Y}) \rightarrow \mathbf{D}(F(\mathcal{X}), F(\mathcal{Y}))$	$\mathbf{C}(\mathcal{X}, \mathcal{Y}) \rightarrow \mathbf{D}(F(\mathcal{X}), F(\mathcal{Y}))$	$\mathbf{C}(\mathcal{X}, \mathcal{Y}) \rightarrow \mathbf{D}(F(\mathcal{X}), F(\mathcal{Y}))$ defined by $f \mapsto F(f)$ is injec-
rph8-g15q_F00238	13	1.000	$f \mapsto F(f)$	$f \mapsto F(f)$	$\mathbf{C}(\mathcal{X}, \mathcal{Y}) \rightarrow \mathbf{D}(F(\mathcal{X}), F(\mathcal{Y}))$ defined by $f \mapsto F(f)$ is injec-
rph8-g15q_F00239	13	0.808	$\mathbf{C} \rightarrow \mathbf{D}$	$\mathbf{C} \rightarrow \mathbf{D}$	Loosely speaking, a functor $\mathbf{C} \rightarrow \mathbf{D}$ is called <i>forgetful</i> if it
rph8-g15q_F00240	14	1.000	$F(\mathcal{X}') = \mathcal{X}$	$F(\mathcal{X}') = \mathcal{X}$	$F(\mathcal{X}') = \mathcal{X}$ and $F(\mathcal{S}(\mathcal{H})) = \mathcal{S}(\mathcal{H})$ are in the image of a
rph8-g15q_F00241	14	1.000	$F(\mathcal{S}(\mathcal{H})') = \mathcal{S}(\mathcal{H})$	$F(\mathcal{S}(\mathcal{H})') = \mathcal{S}(\mathcal{H})$	$F(\mathcal{X}') = \mathcal{X}$ and $F(\mathcal{S}(\mathcal{H})) = \mathcal{S}(\mathcal{H})$ are in the image of a
rph8-g15q_F00242	14	0.880	$F: \mathbf{C} \rightarrow$	$F: \mathbf{C} \rightarrow$	faithful forgetful functor $F: \mathbf{C} \rightarrow \mathbf{Set}$.
rph8-g15q_F00243	14	1.000	$F(\rho') = \rho$	$F(\rho') = \rho$	$\mathbf{C}$ such that $F(\rho') = \rho$. The morphism ρ' is the desired,
rph8-g15q_F00244	14	0.999	$\mathcal{X}' := (\mathcal{X}, \alpha)$	$\mathcal{X}' := (\mathcal{X}, \alpha)$	<i>Example 11.</i> Let $\mathcal{X}' := (\mathcal{X}, \alpha)$ be a data domain that is
rph8-g15q_F00245	14	1.000	$G \ni g \mapsto V_g \in \mathcal{U}(\mathcal{H})$	$G \ni g \mapsto V_g \in \mathcal{U}(\mathcal{H})$	action. Also, if $G \ni g \mapsto V_g \in \mathcal{U}(\mathcal{H})$ is a unitary representa-
rph8-g15q_F00246	14	1.000	$G \ni g \mapsto \mathrm{Ad}_{V_g}$	$G \ni g \mapsto \mathrm{Ad}_{V_g}$	action of G on $\mathcal{S}(\mathcal{H})$ via $G \ni g \mapsto \mathrm{Ad}_{V_g}$, then $\mathcal{S}(\mathcal{H})' :=$
rph8-g15q_F00247	14	1.000	$\mathcal{S}(\mathcal{H})' :=$	$\mathcal{S}(\mathcal{H})' :=$	action of G on $\mathcal{S}(\mathcal{H})$ via $G \ni g \mapsto \mathrm{Ad}_{V_g}$, then $\mathcal{S}(\mathcal{H})' :=$
rph8-g15q_F00248	14	0.902	$\rho': \mathcal{X}' \rightarrow$	$\rho': \mathcal{X}' \rightarrow$	then the quantum encoding “lifts” to a morphism $\rho': \mathcal{X}' \rightarrow$
rph8-g15q_F00249	14	1.000	U	U	the structure preserved under quantum unitary encodings U :
rph8-g15q_F00250	14	1.000	$\mathbb{R}^d \rightarrow \mathcal{U}(\mathcal{H})$	$\mathbb{R}^d \rightarrow \mathcal{U}(\mathcal{H})$	$\mathbb{R}^d \rightarrow \mathcal{U}(\mathcal{H})$ of the form
rph8-g15q_F00251	14	1.000	$\mathcal{L}: \mathbb{R}^d \rightarrow \mathcal{B}(\mathcal{H})$	$\mathcal{L}: \mathbb{R}^d \rightarrow \mathcal{B}(\mathcal{H})$	where $\mathcal{L}: \mathbb{R}^d \rightarrow \mathcal{B}(\mathcal{H})$ is a linear transformation sending
rph8-g15q_F00252	14	1.000	$x \in \mathbb{R}^d$	$x \in \mathbb{R}^d$	each point $x \in \mathbb{R}^d$ to a Hermitian operator $\mathcal{L}(x)$ acting on
rph8-g15q_F00253	14	1.000	$\mathcal{B}(\mathcal{H})$	$\mathcal{B}(\mathcal{H})$	a Hilbert space $\mathcal{H}$ [here, $\mathcal{B}(\mathcal{H})$ denotes the set of bounded
rph8-g15q_F00254	14	1.000	$U: \mathbb{R}^d \rightarrow \mathcal{U}(\mathcal{H})$	$U: \mathbb{R}^d \rightarrow \mathcal{U}(\mathcal{H})$	$U: \mathbb{R}^d \rightarrow \mathcal{U}(\mathcal{H})$ that are morphisms in LieOPS (i.e., that
rph8-g15q_F00255	15	1.000	$\rho: X \rightarrow \mathcal{S}(\mathcal{H})$	$\rho: X \rightarrow \mathcal{S}(\mathcal{H})$	ing $\rho: X \rightarrow \mathcal{S}(\mathcal{H})$ from the space of data points X to the
rph8-g15q_F00256	15	1.000	ρ_{opt}	ρ_{opt}	An optimal solution ρ_{opt} in the context of quantum encodings
rph8-g15q_F00257	16	0.995	$\mathcal{L}: \mathcal{X} \rightarrow \mathcal{B}(\mathcal{H})$	$\mathcal{L}: \mathcal{X} \rightarrow \mathcal{B}(\mathcal{H})$	where $\mathcal{L}: \mathcal{X} \rightarrow \mathcal{B}(\mathcal{H})$ is a linear transformation sending each
rph8-g15q_F00258	16	1.000	$\rho(x) = U(x) \psi_0\rangle \langle \psi_0 U(x)^\dagger$	$\rho(x) = U(x) \psi_0\rangle \langle \psi_0 U(x)^\dagger$	$ \psi_0\rangle \in \mathcal{H}$ via the formula $\rho(x) = U(x) \psi_0\rangle \langle \psi_0 U(x)^\dagger$. Exam-
rph8-g15q_F00259	16	1.000	$\gamma: \mathbb{R} \rightarrow G$	$\gamma: \mathbb{R} \rightarrow G$	morphism $\gamma: \mathbb{R} \rightarrow G$, where $\mathbb{R}$ is viewed as a group under
rph8-g15q_F00260	16	1.000	γ	γ	(1) γ is a continuous function,
rph8-g15q_F00261	16	1.000	$\gamma(0) = 1$	$\gamma(0) = 1$	(2) $\gamma(0) = 1$, and
rph8-g15q_F00262	16	1.000	$\gamma(s+t) = \gamma(s)\gamma(t)$	$\gamma(s+t) = \gamma(s)\gamma(t)$	(3) $\gamma(s+t) = \gamma(s)\gamma(t)$ for all $s, t \in \mathbb{R}$.
rph8-g15q_F00263	16	1.000	$s, t \in \mathbb{R}$	$s, t \in \mathbb{R}$	(3) $\gamma(s+t) = \gamma(s)\gamma(t)$ for all $s, t \in \mathbb{R}$.
rph8-g15q_F00264	16	1.000	$M \in \mathfrak{g}$	$M \in \mathfrak{g}$	Lie group G is generated by a unique element $M \in \mathfrak{g}$ in the

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rph8-g15q_F00265	16	■ 0.999	$\mathfrak{g}$	$\mathfrak{g}$	Lie algebra $\mathfrak{g}$ by the exponential map $\exp : \mathfrak{g} \rightarrow G$. Namely,
rph8-g15q_F00266	16	■ 0.999	$\exp : \mathfrak{g} \rightarrow G$	$\exp : \mathfrak{g} \rightarrow G$	Lie algebra $\mathfrak{g}$ by the exponential map $\exp : \mathfrak{g} \rightarrow G$. Namely,
rph8-g15q_F00267	16	■ 1.000	$t \in \mathbb{R}$	$t \in \mathbb{R}$	for all $t \in \mathbb{R}$. In particular, this implies γ is not only continu-
rph8-g15q_F00268	16	■ 1.000	$t=0$	$t = 0$	γ at $t = 0$ (cf. Sec. 1.3 in Ref. [71]), one obtains $M = \frac{d}{dt}\gamma _{t=0}$.
rph8-g15q_F00269	16	■ 1.000	$M=\left.\frac{d}{dt}\gamma\right _{t=0}$	$M = \left.\frac{d}{dt}\gamma\right _{t=0}$	γ at $t = 0$ (cf. Sec. 1.3 in Ref. [71]), one obtains $M = \frac{d}{dt}\gamma _{t=0}$.
rph8-g15q_F00270	16	■ 1.000	$\mathcal{X}=\mathbb{R}$	$\mathcal{X} = \mathbb{R}$	domain $\mathcal{X}$ of the form $\mathcal{X} = \mathbb{R}$, since in this case $x \mapsto e^{-ixL}$,
rph8-g15q_F00271	16	■ 1.000	$x \mapsto e^{-i x L}$	$x \mapsto e^{-ixL}$	domain $\mathcal{X}$ of the form $\mathcal{X} = \mathbb{R}$, since in this case $x \mapsto e^{-ixL}$,
rph8-g15q_F00272	16	■ 0.999	$L \in \mathfrak{su}(2^n)$	$L \in \mathfrak{su}(2^n)$	with $L \in \mathfrak{su}(2^n)$ for a quantum unitary encoding mapping
rph8-g15q_F00273	16	■ 0.995	$\exp : \mathbb{R}^d \rightarrow \mathbb{R}^d$	$\exp : \mathbb{R}^d \rightarrow \mathbb{R}^d$	noting that the exponential map $\exp : \mathbb{R}^d \rightarrow \mathbb{R}^d$ from the Lie
rph8-g15q_F00274	16	■ 1.000	$\exp (x)=x$	$\exp(x) = x$	$\mathbb{R}^d$ is the identity map, $\exp(x) = x$ for all $x \in \mathbb{R}^d$.
rph8-g15q_F00275	16	■ 1.000	$[V, W]:=V W-W V$	$[V, W] := VW - WV$	$[V, W] := VW - WV$ denotes the commutator.)
rph8-g15q_F00276	16	■ 1.000	H	H	<i>Theorem A1.</i> Let G be any connected Lie group, let H be
rph8-g15q_F00277	16	■ 1.000	$f: G \rightarrow H$	$f : G \rightarrow H$	any Lie group, and let $f : G \rightarrow H$ be a Lie group homomor-
rph8-g15q_F00278	16	■ 0.979	$\mathcal{M}: \mathfrak{g} \rightarrow \mathfrak{h}$	$\mathcal{M} : \mathfrak{g} \rightarrow \mathfrak{h}$	phism. Then, there exists a unique real-linear map $\mathcal{M} : \mathfrak{g} \rightarrow \mathfrak{h}$
rph8-g15q_F00279	16	■ 1.000	$W \in \mathfrak{g}$	$W \in \mathfrak{g}$	for all $W \in \mathfrak{g}$. Moreover, $\mathcal{M}$ is a Lie algebra homomorphism,
rph8-g15q_F00280	16	■ 1.000	$\mathcal{M}$	$\mathcal{M}$	for all $W \in \mathfrak{g}$. Moreover, $\mathcal{M}$ is a Lie algebra homomorphism,
rph8-g15q_F00281	16	■ 1.000	$W_1, W_2 \in \mathfrak{g}$	$W_1, W_2 \in \mathfrak{g}$	for all $W_1, W_2 \in \mathfrak{g}$.
rph8-g15q_F00282	17	■ 1.000	$U: \mathcal{X} \rightarrow \mathcal{U}(\mathbb{C}^2)$	$U : \mathcal{X} \rightarrow \mathcal{U}(\mathbb{C}^2)$	$\mathcal{X} = \mathbb{R}^2$. Suppose that $U : \mathcal{X} \rightarrow \mathcal{U}(\mathbb{C}^2)$ is a quantum unitary
rph8-g15q_F00283	17	■ 1.000	$(s, t) \in \mathbb{R}^2$	$(s, t) \in \mathbb{R}^2$	for all $(s, t) \in \mathbb{R}^2$. Then, this defines a perfectly reasonable
rph8-g15q_F00284	17	■ 1.000	$\mathcal{M}: \mathbb{R}^2 \rightarrow \mathfrak{su}(2)$	$\mathcal{M} : \mathbb{R}^2 \rightarrow \mathfrak{su}(2)$	sociated with Eq. (A5). First, let $\mathcal{M} : \mathbb{R}^2 \rightarrow \mathfrak{su}(2)$ be the
rph8-g15q_F00285	17	■ 1.000	$\mathcal{M}((1,0))=-i X$	$\mathcal{M}((1,0)) = -iX$	linear map uniquely determined by $\mathcal{M}((1,0)) = -iX$ and
rph8-g15q_F00286	17	■ 1.000	$\mathcal{M}((0,1))=-i Y$	$\mathcal{M}((0,1)) = -iY$	$\mathcal{M}((0,1)) = -iY$, so that $\mathcal{M}((s,t)) = -isX - itY$ for all
rph8-g15q_F00287	17	■ 1.000	$\mathcal{M}((s, t))=-i s X-i t Y$	$\mathcal{M}((s,t)) = -isX - itY$	$\mathcal{M}((0,1)) = -iY$, so that $\mathcal{M}((s,t)) = -isX - itY$ for all
rph8-g15q_F00288	17	■ 0.968	$\exp : \mathbb{R}^2 \rightarrow \mathbb{R}^2$	$\exp : \mathbb{R}^2 \rightarrow \mathbb{R}^2$	$(s, t) \in \mathbb{R}^2$. Since the exponential map $\exp : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ going
rph8-g15q_F00289	17	■ 1.000	$\mathbb{R}^2$	$\mathbb{R}^2$	from the Lie algebra of the Lie group $\mathbb{R}^2$ (with group struc-
rph8-g15q_F00290	17	■ 0.995	$\mathbb{R}_\alpha \subset \mathbb{R}^2$	$\mathbb{R}_\alpha \subset \mathbb{R}^2$	$\mathbb{R}_\alpha \subset \mathbb{R}^2$ generated by some nonzero vector α , i.e.,
rph8-g15q_F00291	17	■ 1.000	$U _{\mathbb{R}_\alpha}: \mathbb{R}_\alpha \rightarrow \mathcal{U}(\mathbb{C}^2)$	$U _{\mathbb{R}_\alpha} : \mathbb{R}_\alpha \rightarrow \mathcal{U}(\mathbb{C}^2)$	group homomorphism $U _{\mathbb{R}_\alpha} : \mathbb{R}_\alpha \rightarrow \mathcal{U}(\mathbb{C}^2)$ since
rph8-g15q_F00292	17	■ 1.000	$W:=\alpha \cdot(X, Y)$	$W := \alpha \cdot (X, Y)$	where $W := \alpha \cdot (X, Y)$ is the dot product of α with the vector
rph8-g15q_F00293	17	■ 0.993	X, Y	X, Y	of operators (X, Y) . The second equality in Eq. (A8) holds
rph8-g15q_F00294	17	■ 1.000	W	W	because every matrix W commutes with itself. In other words,
rph8-g15q_F00295	17	■ 1.000	$\mathbb{R} \stackrel{\alpha}{\rightarrow} \mathbb{R}^2$	$\mathbb{R} \stackrel{\alpha}{\rightarrow} \mathbb{R}^2$	the one-parameter subgroup $\mathbb{R} \stackrel{\alpha}{\rightarrow} \mathbb{R}^2$ of $\mathbb{R}^2$ sending $r \in \mathbb{R}$
rph8-g15q_F00296	17	■ 1.000	$r \in \mathbb{R}$	$r \in \mathbb{R}$	the one-parameter subgroup $\mathbb{R} \stackrel{\alpha}{\rightarrow} \mathbb{R}^2$ of $\mathbb{R}^2$ sending $r \in \mathbb{R}$
rph8-g15q_F00297	17	■ 1.000	$\alpha(r):=r \alpha$	$\alpha(r) := r\alpha$	to $\alpha(r) := r\alpha$ is a one-parameter subgroup of $\mathbb{R}^2$ that gets
rph8-g15q_F00298	17	■ 1.000	$\mathcal{U}(\mathbb{C}^2)$	$\mathcal{U}(\mathbb{C}^2)$	of $\mathcal{U}(\mathbb{C}^2)$ that sends $r \in \mathbb{R}$ to $U(r\alpha)$ [126].
rph8-g15q_F00299	17	■ 1.000	$U(r \alpha)$	$U(r\alpha)$	of $\mathcal{U}(\mathbb{C}^2)$ that sends $r \in \mathbb{R}$ to $U(r\alpha)$ [126].
rph8-g15q_F00300	17	■ 1.000	$f: G \rightarrow$	$f : G \rightarrow$	whose objects are Lie groups and where a morphism $f : G \rightarrow$
rph8-g15q_F00301	17	■ 1.000	$\mathbb{R} \stackrel{\gamma}{\rightarrow} G$	$\mathbb{R} \stackrel{\gamma}{\rightarrow} G$	every one-parameter subgroup $\mathbb{R} \stackrel{\gamma}{\rightarrow} G$ in G , the composite
rph8-g15q_F00302	17	■ 1.000	$\mathbb{R} \stackrel{\gamma}{\rightarrow} G \stackrel{f}{\rightarrow} H$	$\mathbb{R} \stackrel{\gamma}{\rightarrow} G \stackrel{f}{\rightarrow} H$	$\mathbb{R} \stackrel{\gamma}{\rightarrow} G \stackrel{f}{\rightarrow} H$ is a one-parameter subgroup in H .
rph8-g15q_F00303	17	■ 1.000	$G \stackrel{f}{\rightarrow} H$	$G \stackrel{f}{\rightarrow} H$	<i>Theorem A2.</i> Let $G \stackrel{f}{\rightarrow} H$ be an OPS homomorphism of
rph8-g15q_F00304	17	■ 1.000	$x \in \mathfrak{g}$	$x \in \mathfrak{g}$	for all $x \in \mathfrak{g}$.
rph8-g15q_F00305	17	■ 1.000	$1_G \in G$	$1_G \in G$	called the <i>pushforward</i>) of f at the identity $1_G \in G$, i.e., $\mathcal{M} :=$
rph8-g15q_F00306	17	■ 1.000	$\mathcal{M}:=$	$\mathcal{M} :=$	called the <i>pushforward</i>) of f at the identity $1_G \in G$, i.e., $\mathcal{M} :=$
rph8-g15q_F00307	17	■ 1.000	$D f _{1_G}$	$Df _{1_G}$	$Df _{1_G}$. To be somewhat self-contained, let us recall how $\mathcal{M}$
rph8-g15q_F00308	17	■ 1.000	$\gamma(0)=1_G$	$\gamma(0) = 1_G$	groups). For any smooth curve $\gamma : \mathbb{R} \rightarrow G$ with $\gamma(0) = 1_G$
rph8-g15q_F00309	17	■ 1.000	$\left.\frac{d}{dt}\gamma\right _{t=0}=x \in \mathfrak{g}$	$\left.\frac{d}{dt}\gamma\right _{t=0} = x \in \mathfrak{g}$	and $\frac{d}{dt}\gamma _{t=0} = x \in \mathfrak{g}$, we have $\frac{d}{dt}(f \circ \gamma) _{t=0} = \mathcal{M}(x)$. The

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rph8-g15q_F00310	17	■ 1.000	$\left.\frac{d}{dt}(f \circ \gamma)\right _{t=0} = \mathcal{M}(x)$	$\left.\frac{d}{dt}(f \circ \gamma)\right _{t=0} = \mathcal{M}(x)$	and $\left.\frac{d}{dt}\gamma\right _{t=0} = x \in \mathfrak{g}$, we have $\left.\frac{d}{dt}(f \circ \gamma)\right _{t=0} = \mathcal{M}(x)$. The
rph8-g15q_F00311	17	■ 1.000	$\left.\frac{d}{dt}\gamma\right _{t=0} = x$	$\left.\frac{d}{dt}\gamma\right _{t=0} = x$	$\left.\frac{d}{dt}\gamma\right _{t=0} = x$. By Eq. (A2) for γ , this means that
rph8-g15q_F00312	17	■ 1.000	$f \circ \gamma$	$f \circ \gamma$	by Eq. (A3) for $f \circ \gamma$, there exists a unique element $y \in \mathfrak{h}$
rph8-g15q_F00313	17	■ 1.000	$y \in \mathfrak{h}$	$y \in \mathfrak{h}$	by Eq. (A3) for $f \circ \gamma$, there exists a unique element $y \in \mathfrak{h}$
rph8-g15q_F00314	18	■ 1.000	$G = \mathbb{R}^d$	$G = \mathbb{R}^d$	unitary encodings by taking $G = \mathbb{R}^d$ to be the additive group
rph8-g15q_F00315	18	■ 1.000	$\mathfrak{g} = \mathbb{R}^d$	$\mathfrak{g} = \mathbb{R}^d$	and $\mathfrak{g} = \mathbb{R}^d$ to be its associated Lie algebra. Note that in this
rph8-g15q_F00316	18	■ 1.000	$\mathbb{R}^d \xrightarrow{f} G$	$\mathbb{R}^d \xrightarrow{f} G$	<i>Corollary A1.</i> Let $\mathbb{R}^d \xrightarrow{f} G$ be an OPS homomorphism of
rph8-g15q_F00317	18	■ 0.976	$\mathcal{M} : \mathbb{R}^d \rightarrow$	$\mathcal{M} : \mathbb{R}^d \rightarrow$	Lie groups. Then, there exists a unique linear map $\mathcal{M} : \mathbb{R}^d \rightarrow$
rph8-g15q_F00318	18	■ 1.000	$f : \mathbb{R}^d \rightarrow G$	$f : \mathbb{R}^d \rightarrow G$	$\mathfrak{g}$, the function $f : \mathbb{R}^d \rightarrow G$ specified by Eq. (A15) defines an
rph8-g15q_F00319	18	■ 1.000	$\mathbb{R}^d \rightarrow \mathcal{U}(\mathbb{C}^n)$	$\mathbb{R}^d \rightarrow \mathcal{U}(\mathbb{C}^n)$	$\mathbb{R}^d \rightarrow \mathcal{U}(\mathbb{C}^n)$ of the form
rph8-g15q_F00320	18	■ 0.999	$\mathcal{L} : \mathbb{R}^d \rightarrow \mathcal{B}(\mathbb{C}^n)$	$\mathcal{L} : \mathbb{R}^d \rightarrow \mathcal{B}(\mathbb{C}^n)$	where $\mathcal{L} : \mathbb{R}^d \rightarrow \mathcal{B}(\mathbb{C}^n)$ is a linear map such that $\mathcal{L}(x)$ is self-
rph8-g15q_F00321	18	■ 1.000	$\mathbb{R}^d \xrightarrow{U} \mathcal{U}(\mathbb{C}^n)$	$\mathbb{R}^d \xrightarrow{U} \mathcal{U}(\mathbb{C}^n)$	adjoint for each $x \in \mathbb{R}^d$, defines a morphism $\mathbb{R}^d \xrightarrow{U} \mathcal{U}(\mathbb{C}^n)$
rph8-g15q_F00322	18	■ 1.000	$U(x+y) = e^{-i\mathcal{L}(x+y)}$	$U(x+y) = e^{-i\mathcal{L}(x+y)}$	general. This is because $U(x+y) = e^{-i\mathcal{L}(x+y)}$ is <i>not</i> in gen-
rph8-g15q_F00323	18	■ 1.000	$U(x)U(y) = e^{-i\mathcal{L}(x)}e^{-i\mathcal{L}(y)}$	$U(x)U(y) = e^{-i\mathcal{L}(x)}e^{-i\mathcal{L}(y)}$	eral equal to $U(x)U(y) = e^{-i\mathcal{L}(x)}e^{-i\mathcal{L}(y)}$. Because the group
rph8-g15q_F00324	18	■ 1.000	$X \subseteq \mathbb{R}^2$	$X \subseteq \mathbb{R}^2$	a dataset $X \subseteq \mathbb{R}^2$. The labeling is determined by a binary
rph8-g15q_F00325	18	■ 1.000	$c : \mathcal{X} \rightarrow \{-1, 0, +1\}$	$c : \mathcal{X} \rightarrow \{-1, 0, +1\}$	<i>classifier</i> , which is a function $c : \mathcal{X} \rightarrow \{-1, 0, +1\}$, where
rph8-g15q_F00326	18	■ 1.000	$O \in$	$O \in$	$\rho : \mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$ if and only if there exists an observable $O \in$
rph8-g15q_F00327	18	■ 0.997	O	O	$\mathcal{B}(\mathcal{H})$ such that the spectrum of O is $\{-1, +1\}$ and
rph8-g15q_F00328	18	■ 0.997	$\{-1, +1\}$	$\{-1, +1\}$	$\mathcal{B}(\mathcal{H})$ such that the spectrum of O is $\{-1, +1\}$ and
rph8-g15q_F00329	18	■ 1.000	$\rho(x)$	$\rho(x)$	which is the expectation value of O in the state $\rho(x)$. The func-
rph8-g15q_F00330	18	■ 1.000	y	y	tional y is called the <i>quantum classifier</i> and the set $y^{-1}(0) \subseteq \mathcal{X}$
rph8-g15q_F00331	18	■ 1.000	$y^{-1}(0) \subseteq \mathcal{X}$	$y^{-1}(0) \subseteq \mathcal{X}$	tional y is called the <i>quantum classifier</i> and the set $y^{-1}(0) \subseteq \mathcal{X}$
rph8-g15q_F00332	18	■ 1.000	$x_1, x_2 \in \mathbb{R}$	$x_1, x_2 \in \mathbb{R}$	for all $x_1, x_2 \in \mathbb{R}$. Note that invariance of the quantum clas-
rph8-g15q_F00333	18	■ 1.000	$ \psi_0\rangle \in \mathcal{H} = \mathbb{C}^4$	$ \psi_0\rangle \in \mathcal{H} = \mathbb{C}^4$	Now let $ \psi_0\rangle \in \mathcal{H} = \mathbb{C}^4$ be the initial state
rph8-g15q_F00334	18	■ 1.000	$p = 0.99$	$p = 0.99$	with $p = 0.99$, let O be the observable
rph8-g15q_F00335	18	■ 0.969	(x_1, x_2)	(x_1, x_2)	The orange region in Fig. 3 consists of the points (x_1, x_2) such
rph8-g15q_F00336	18	■ 1.000	$y(x_1, x_2) < 0$	$y(x_1, x_2) < 0$	that $y(x_1, x_2) < 0$, while the blue region consists of the points
rph8-g15q_F00337	18	■ 0.999	$y(x_1, x_2) > 0$	$y(x_1, x_2) > 0$	(x_1, x_2) such that $y(x_1, x_2) > 0$. The boundary between the
rph8-g15q_F00338	18	■ 0.965	$y(x_1, x_2) =$	$y(x_1, x_2) =$	two regions consists of the points (x_1, x_2) such that $y(x_1, x_2) =$
rph8-g15q_F00339	18	■ 1.000	$U : \mathbb{R}^2 \rightarrow \mathcal{U}(\mathbb{C}^4)$	$U : \mathbb{R}^2 \rightarrow \mathcal{U}(\mathbb{C}^4)$	encodings $\hat{U} : \mathbb{R}^2 \rightarrow \mathcal{U}(\mathbb{C}^4)$ are of the form $U(x) = e^{-i\mathcal{L}(x)}$
rph8-g15q_F00340	18	■ 1.000	$U(x) = e^{-i\mathcal{L}(x)}$	$U(x) = e^{-i\mathcal{L}(x)}$	encodings $\hat{U} : \mathbb{R}^2 \rightarrow \mathcal{U}(\mathbb{C}^4)$ are of the form $U(x) = e^{-i\mathcal{L}(x)}$
rph8-g15q_F00341	18	■ 1.000	$\mathcal{L} : \mathbb{R}^2 \rightarrow \mathfrak{su}(4)$	$\mathcal{L} : \mathbb{R}^2 \rightarrow \mathfrak{su}(4)$	for some linear transformation $\mathcal{L} : \mathbb{R}^2 \rightarrow \mathfrak{su}(4)$ as in Eq. (14).
rph8-g15q_F00342	18	■ 1.000	e_1, e_2	e_1, e_2	the value of $\mathcal{L}$ on the basis elements e_1, e_2 of $\mathbb{R}^2$. Namely,
rph8-g15q_F00343	18	■ 1.000	$L_1 := \mathcal{L}(e_1)$	$L_1 := \mathcal{L}(e_1)$	$L_1 := \mathcal{L}(e_1)$ and $L_2 := \mathcal{L}(e_2)$. The fact that L_1 is a traceless
rph8-g15q_F00344	18	■ 1.000	$L_2 := \mathcal{L}(e_2)$	$L_2 := \mathcal{L}(e_2)$	$L_1 := \mathcal{L}(e_1)$ and $L_2 := \mathcal{L}(e_2)$. The fact that L_1 is a traceless
rph8-g15q_F00345	18	■ 1.000	$a_{11}, a_{22}, a_{33}, a_{44} \in \mathbb{R}, a_{12}, a_{13}, a_{14}, a_{23}, a_{24}, a_{34} \in \mathbb{C}$	$a_{11}, a_{22}, a_{33}, a_{44} \in \mathbb{R}, a_{12}, a_{13}, a_{14}, a_{23}, a_{24}, a_{34} \in \mathbb{C}$	where $a_{11}, a_{22}, a_{33}, a_{44} \in \mathbb{R}, a_{12}, a_{13}, a_{14}, a_{23}, a_{24}, a_{34} \in \mathbb{C}$,
rph8-g15q_F00346	18	■ 1.000	$a_{11} + a_{22} + a_{33} + a_{44} = 0$	$a_{11} + a_{22} + a_{33} + a_{44} = 0$	and $a_{11} + a_{22} + a_{33} + a_{44} = 0$, and similarly for the matrix
rph8-g15q_F00347	18	■ 0.994	L_2	L_2	L_2 . This parametrization leads to a $2(4 + 2(6) - 1) = 30$ -
rph8-g15q_F00348	18	■ 0.994	$2(4 + 2(6) - 1) = 30$	$2(4 + 2(6) - 1) = 30$	L_2 . This parametrization leads to a $2(4 + 2(6) - 1) = 30$ -
rph8-g15q_F00349	18	■ 1.000	$\mathbb{R}^2 \rightarrow \mathcal{U}(\mathbb{C}^4)$	$\mathbb{R}^2 \rightarrow \mathcal{U}(\mathbb{C}^4)$	$\mathbb{R}^2 \rightarrow \mathcal{U}(\mathbb{C}^4)$ that are of the form $\hat{U}(x) = e^{-i\mathcal{L}(x)}$ for some
rph8-g15q_F00350	18	■ 0.989	$L_2 =$	$L_2 =$	The equivariance constraint $L_2 = \text{SWAP } L_1 \text{ SWAP}$ from Eq.

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rph8-g15q_F00351	19	■ 1.000	$\left\{X \otimes X, L_1\right\}=0$	$\{X \otimes X, L_1\} = 0$	$\{X \otimes X, L_1\} = 0$ from Eq. (18) yields
rph8-g15q_F00352	19	■ 1.000	$\mathbb{R}^n$	$\mathbb{R}^n$	cussion on metric spaces that have underlying sets $\mathbb{R}^n$ and
rph8-g15q_F00353	19	■ 0.936	$\mathcal{X}=\mathbb{R}^n$	$\mathcal{X} = \mathbb{R}^n$	In more detail, let $\mathcal{X} = \mathbb{R}^n$ and equip it with a Mahalanobis
rph8-g15q_F00354	19	■ 0.432	$\left(\mathcal{Y}=\mathbb{R}^m, d_Y\right)$	$(\mathcal{Y} = \mathbb{R}^m, d_Y)$	distance metric, where $(\mathcal{Y} = \mathbb{R}^m, d_Y)$ is a Euclidean space
rph8-g15q_F00355	19	■ 1.000	$\mathbb{R}^n \xrightarrow{\varphi} \mathbb{R}^m$	$\mathbb{R}^n \xrightarrow{\varphi} \mathbb{R}^m$	(so that d_Y is the Euclidean distance) and $\mathbb{R}^n \xrightarrow{\varphi} \mathbb{R}^m$ is an
rph8-g15q_F00356	19	■ 1.000	$m \times n$	$m \times n$	injective linear map, described by some $m \times n$ matrix V with
rph8-g15q_F00357	19	■ 1.000	V	V	injective linear map, described by some $m \times n$ matrix V with
rph8-g15q_F00358	19	■ 1.000	V^T	V^T	where V^T denotes the transpose of V . Although this exam-
rph8-g15q_F00359	19	■ 0.871	$\left(\mathbb{R}^m, A\right)$	$(\mathbb{R}^m, A)$	$(\mathbb{R}^m, A)$, with $m \in \mathbb{N}$ and A a positive-definite $m \times m$ matrix.
rph8-g15q_F00360	19	■ 0.871	$m \in \mathbb{N}$	$m \in \mathbb{N}$	$(\mathbb{R}^m, A)$, with $m \in \mathbb{N}$ and A a positive-definite $m \times m$ matrix.
rph8-g15q_F00361	19	■ 0.871	A	A	$(\mathbb{R}^m, A)$, with $m \in \mathbb{N}$ and A a positive-definite $m \times m$ matrix.
rph8-g15q_F00362	19	■ 0.871	$m \times m$	$m \times m$	$(\mathbb{R}^m, A)$, with $m \in \mathbb{N}$ and A a positive-definite $m \times m$ matrix.
rph8-g15q_F00363	19	■ 0.945	$\mathbb{R}^m, A$	$\mathbb{R}^m, A$	A morphism in Mahal from $(\mathbb{R}^m, A)$ to $(\mathbb{R}^n, B)$ is an $n \times m$
rph8-g15q_F00364	19	■ 0.945	$\mathbb{R}^n, B$	$\mathbb{R}^n, B$	A morphism in Mahal from $(\mathbb{R}^m, A)$ to $(\mathbb{R}^n, B)$ is an $n \times m$
rph8-g15q_F00365	19	■ 0.945	$n \times m$	$n \times m$	A morphism in Mahal from $(\mathbb{R}^m, A)$ to $(\mathbb{R}^n, B)$ is an $n \times m$
rph8-g15q_F00366	19	■ 1.000	$V^T B V=A$	$V^T B V = A$	matrix V with trivial kernel such that $V^T B V = A$. The trivial
rph8-g15q_F00367	19	■ 1.000	$\mathbb{R}^m$	$\mathbb{R}^m$	$\mathbb{R}^m$ to $\mathbb{R}^n$. The category Mahal is <i>equivalent</i> [130] to the
rph8-g15q_F00368	19	■ 1.000	$\mathbb{R}^m \xrightarrow{V} \mathbb{R}^n$	$\mathbb{R}^m \xrightarrow{V} \mathbb{R}^n$	the task can be described as finding a linear map $\mathbb{R}^m \xrightarrow{V} \mathbb{R}^n$,
rph8-g15q_F00369	19	■ 1.000	$y \in \mathcal{Y}$	$y \in \mathcal{Y}$	holds for all $y \in \mathcal{Y}$. In particular, for any $x_3 \in \mathcal{X}$,
rph8-g15q_F00370	19	■ 1.000	$x_3 \in \mathcal{X}$	$x_3 \in \mathcal{X}$	holds for all $y \in \mathcal{Y}$. In particular, for any $x_3 \in \mathcal{X}$,
rph8-g15q_F00371	19	■ 1.000	$d_{\mathcal{X}}(x, x)=0$	$d_{\mathcal{X}}(x, x) = 0$	true that $d_{\mathcal{X}}(x, x) = 0$ for all $x \in \mathcal{X}$.
rph8-g15q_F00372	19	■ 1.000	$f\left(x_1\right)=f\left(x_2\right)$	$f\left(x_1\right)=f\left(x_2\right)$	then $f(x_1) = f(x_2)$. Hence, f is one to one if and only
rph8-g15q_F00373	19	■ 1.000	x_1, x_2	x_1, x_2	if $d_{\mathcal{X}}(x_1, x_2) = 0$ implies $x_1 = x_2$ for all x_1, x_2 that satisfy
rph8-g15q_F00374	19	■ 1.000	$X \subset \mathcal{X}$	$X \subset \mathcal{X}$	classes. Although the given data $X \subset \mathcal{X}$ shown in Fig. 8 are
rph8-g15q_F00375	20	■ 1.000	$\{A, B\}$	$\{A, B\}$	Let $\{A, B\}$ be a partition of X , where the elements of A are
rph8-g15q_F00376	20	■ 0.997	B	B	labeled as class +1 and the elements of B are labeled as class
rph8-g15q_F00377	20	■ 0.954	$c: \mathcal{X} \rightarrow\{-1,0,1\}$	$c: \mathcal{X} \rightarrow \{-1, 0, 1\}$	−1. The associated classifier $c: \mathcal{X} \rightarrow \{-1, 0, 1\}$ satisfies
rph8-g15q_F00378	20	■ 0.965	$\# A$	$\# A$	where $\#A$ and $\#B$ denote the cardinalities of A and B , respec-
rph8-g15q_F00379	20	■ 0.965	$\# B$	$\# B$	where $\#A$ and $\#B$ denote the cardinalities of A and B , respec-
rph8-g15q_F00380	20	■ 1.000	$\mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$	$\mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$	not a trivial task because the space of encodings $\mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$
rph8-g15q_F00381	20	■ 1.000	$\rho: \Theta \times \mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$	$\rho: \Theta \times \mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$	a <i>parametrized</i> family of encodings $\rho: \Theta \times \mathcal{X} \rightarrow \mathcal{S}(\mathcal{H})$,
rph8-g15q_F00382	20	■ 1.000	$\Theta=\left[0,2 \pi\right]^4$	$\Theta = [0, 2\pi]^4$	this work.) More specifically, we take $\Theta = [0, 2\pi]^4$ and write
rph8-g15q_F00383	20	■ 1.000	$\theta=\left(\theta_1, \theta_2, \theta_3, \theta_4\right) \in \Theta$	$\theta = (\theta_1, \theta_2, \theta_3, \theta_4) \in \Theta$	$\theta = (\theta_1, \theta_2, \theta_3, \theta_4) \in \Theta$ and $\rho_{\theta}(x) := \rho(\theta, x)$. Moreover, we
rph8-g15q_F00384	20	■ 1.000	$\rho_{\theta}(x):=\rho(\theta, x)$	$\rho_{\theta}(x) := \rho(\theta, x)$	$\theta = (\theta_1, \theta_2, \theta_3, \theta_4) \in \Theta$ and $\rho_{\theta}(x) := \rho(\theta, x)$. Moreover, we
rph8-g15q_F00385	20	■ 1.000	$U: \Theta \times \mathcal{X} \rightarrow \mathcal{U}(\mathcal{H})$	$U: \Theta \times \mathcal{X} \rightarrow \mathcal{U}(\mathcal{H})$	$U: \Theta \times \mathcal{X} \rightarrow \mathcal{U}(\mathcal{H})$, where
rph8-g15q_F00386	20	■ 1.000	ρ_A	ρ_A	which is an observable that depends on θ (since ρ_A and ρ_B
rph8-g15q_F00387	20	■ 1.000	ρ_B	ρ_B	which is an observable that depends on θ (since ρ_A and ρ_B
rph8-g15q_F00388	20	■ 0.999	$\lambda \in(0,1]$	$\lambda \in (0, 1]$	matrices. Then, there exists a $\lambda \in (0, 1]$ such that the eigen-
rph8-g15q_F00389	20	■ 1.000	$\rho_A-\rho_B$	$\rho_A - \rho_B$	values of $\rho_A - \rho_B$ are $\{-\lambda, +\lambda\}$.
rph8-g15q_F00390	20	■ 1.000	$\{-\lambda,+\lambda\}$	$\{-\lambda, +\lambda\}$	values of $\rho_A - \rho_B$ are $\{-\lambda, +\lambda\}$.
rph8-g15q_F00391	20	■ 1.000	λ_1	λ_1	λ_1 and λ_2 denote the two eigenvalues of $\rho_A - \rho_B$. Since the
rph8-g15q_F00392	20	■ 1.000	λ_2	λ_2	λ_1 and λ_2 denote the two eigenvalues of $\rho_A - \rho_B$. Since the
rph8-g15q_F00393	20	■ 0.955	$\rho_A \neq \rho_B$	$\rho_A \neq \rho_B$	of $\rho_A - \rho_B$ are between −1 and 1. Since $\rho_A \neq \rho_B$, at least one
rph8-g15q_F00394	20	■ 0.982	$\lambda_1+\lambda_2=0$	$\lambda_1 + \lambda_2 = 0$	i.e., $\lambda_1 + \lambda_2 = 0$. In other words, $\lambda_2 = -\lambda_1$. Setting $\lambda = \lambda_1 $
rph8-g15q_F00395	20	■ 0.982	$\lambda_2=-\lambda_1$	$\lambda_2 = -\lambda_1$	i.e., $\lambda_1 + \lambda_2 = 0$. In other words, $\lambda_2 = -\lambda_1$. Setting $\lambda = \lambda_1 $
rph8-g15q_F00396	20	■ 0.982	$\lambda=\left \lambda_1\right $	$\lambda = \lambda_1 $	i.e., $\lambda_1 + \lambda_2 = 0$. In other words, $\lambda_2 = -\lambda_1$. Setting $\lambda = \lambda_1 $
rph8-g15q_F00397	20	■ 1.000	$A \cup B=X$	$A \cup B = X$	gether with a partition $A \cup B = X$, with elements of A and B
rph8-g15q_F00398	20	■ 1.000	$O=\rho_A-\rho_B$	$O = \rho_A - \rho_B$	Eq. (D6), are distinct density matrices. Set $O = \rho_A - \rho_B$ and
rph8-g15q_F00399	20	■ 1.000	$\lambda>0$	$\lambda > 0$	let $\{-\lambda, +\lambda\}$ denote the set of eigenvalues of O , with $\lambda > 0$.

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rph8-g15q_F00400	20	■ 1.000	c	c	Finally, let c be the classifier in Eq. (D5) and let $y : \mathcal{X} \rightarrow \mathbb{R}$
rph8-g15q_F00401	20	■ 0.856	$x \in X$	$x \in X$	for all $x \in X$, where $\text{sgn} : (-\infty, 0) \cup (0, \infty) \rightarrow \{-1, +1\}$ de-
rph8-g15q_F00402	20	■ 0.856	$\operatorname{sgn}:(-\infty, 0) \cup (0, \infty) \rightarrow \{-1, +1\}$	$\text{sgn} : (-\infty, 0) \cup (0, \infty) \rightarrow \{-1, +1\}$	for all $x \in X$, where $\text{sgn} : (-\infty, 0) \cup (0, \infty) \rightarrow \{-1, +1\}$ de-
rph8-g15q_F00403	20	■ 1.000	$\operatorname{sgn}(x)=\frac{x}{ x }$	$\text{sgn}(x) = \frac{x}{ x }$	notes the sign function $\text{sgn}(x) = \frac{x}{ x }$.
rph8-g15q_F00404	20	■ 1.000	$\Pi_+-\Pi_-$	$\Pi_+ - \Pi_-$	observable $\Pi_+ - \Pi_-$, where Π_+ and Π_- are the projections
rph8-g15q_F00405	20	■ 1.000	Π_+	Π_+	observable $\Pi_+ - \Pi_-$, where Π_+ and Π_- are the projections
rph8-g15q_F00406	20	■ 1.000	Π_-	Π_-	observable $\Pi_+ - \Pi_-$, where Π_+ and Π_- are the projections
rph8-g15q_F00407	20	■ 1.000	$\theta_1, \theta_2, \theta_3$	$\theta_1, \theta_2, \theta_3$	we obtain specific values of $\theta_1, \theta_2, \theta_3$, and θ_4 for the
rph8-g15q_F00408	20	■ 1.000	θ_4	θ_4	we obtain specific values of $\theta_1, \theta_2, \theta_3$, and θ_4 for the
rph8-g15q_F00409	20	■ 1.000	$\rho:\Theta\times\mathcal{X}\rightarrow\mathscr{S}(\mathbb{C}^2)$	$\rho : \Theta \times \mathcal{X} \rightarrow \mathscr{S}(\mathbb{C}^2)$	parametrized encoding $\rho : \Theta \times \mathcal{X} \rightarrow \mathscr{S}(\mathbb{C}^2)$. The associated
rph8-g15q_F00410	20	■ 1.000	$\mathcal{X}\times\mathcal{X}$	$\mathcal{X} \times \mathcal{X}$	the distance between points on a discretized grid from $\mathcal{X} \times \mathcal{X}$
rph8-g15q_F00411	21	■ 1.000	$G:\mathbf{D}\rightarrow\mathbf{E}$	$G : \mathbf{D} \rightarrow \mathbf{E}$	<i>Definition E1.</i> Let $F : \mathbf{C} \rightarrow \mathbf{D}$ and $G : \mathbf{D} \rightarrow \mathbf{E}$ be two
rph8-g15q_F00412	21	■ 1.000	$G\circ F:\mathbf{C}\rightarrow\mathbf{E}$	$G \circ F : \mathbf{C} \rightarrow \mathbf{E}$	functors. The <i>composite</i> $G \circ F : \mathbf{C} \rightarrow \mathbf{E}$ is the functor that
rph8-g15q_F00413	21	■ 1.000	$G(F(\mathcal{X}))$	$G(F(\mathcal{X}))$	sends each object $\mathcal{X}$ in $\mathbf{C}$ to $G(F(\mathcal{X}))$ and sends each mor-
rph8-g15q_F00414	21	■ 0.999	$G(F(f))$	$G(F(f))$	phism $f : \mathcal{X} \rightarrow \mathcal{Y}$ in $\mathbf{C}$ to the morphism $G(F(f))$.
rph8-g15q_F00415	21	■ 0.989	F, G	F, G	<i>Definition E2.</i> Let $\mathbf{C}$ and $\mathbf{D}$ be categories, and let $F, G :$
rph8-g15q_F00416	21	■ 0.996	η	η	$\mathbf{C} \rightarrow \mathbf{D}$ be two functors. A <i>natural transformation</i> η from F
rph8-g15q_F00417	21	■ 1.000	$\eta:F\Rightarrow G$	$\eta : F \Rightarrow G$	to G , written $\eta : F \Rightarrow G$, associates with each object $\mathcal{X}$ in $\mathbf{C}$
rph8-g15q_F00418	21	■ 0.948	$\eta_{\mathcal{X}}:F(\mathcal{X})\rightarrow G(\mathcal{X})$	$\eta_{\mathcal{X}} : F(\mathcal{X}) \rightarrow G(\mathcal{X})$	a morphism $\eta_{\mathcal{X}} : F(\mathcal{X}) \rightarrow G(\mathcal{X})$ in $\mathbf{D}$ such that the diagram
rph8-g15q_F00419	21	■ 0.996	$G(f)\circ\eta_{\mathcal{X}}=\eta_{\mathcal{Y}}\circ F(f)$	$G(f) \circ \eta_{\mathcal{X}} = \eta_{\mathcal{Y}} \circ F(f)$	in $\mathbf{D}$ commutes, i.e., $G(f) \circ \eta_{\mathcal{X}} = \eta_{\mathcal{Y}} \circ F(f)$, for every mor-
rph8-g15q_F00420	21	■ 1.000	$\mathcal{C}$	$\mathcal{C}$	phism $\mathcal{X} \xrightarrow{f} \mathcal{Y}$ in $\mathcal{C}$. The commutativity of diagram (E1) is
rph8-g15q_F00421	21	■ 1.000	$\mathbb{C}$	$\mathbb{C}$	phisms are $\mathbb{C}$ -linear transformations. Moreover, there is a
rph8-g15q_F00422	21	■ 1.000	$(\mathbb{C}_{\text{fs}}^{\mathcal{X}}, 0, +, \cdot)$	$(\mathbb{C}_{\text{fs}}^{\mathcal{X}}, 0, +, \cdot)$	a set $\mathcal{X}$ the vector space $(\mathbb{C}_{\text{fs}}^{\mathcal{X}}, 0, +, \cdot)$ generated by it, which
rph8-g15q_F00423	21	■ 1.000	$\mathbb{C}_{\text{fs}}^{\mathcal{X}}\rightarrow\mathbb{C}_{\text{fs}}^{\mathcal{Y}}$	$\mathbb{C}_{\text{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\text{fs}}^{\mathcal{Y}}$	$\mathcal{Y}$ a linear transformation $\mathbb{C}_{\text{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\text{fs}}^{\mathcal{Y}}$. This passage defines
rph8-g15q_F00424	21	■ 0.995	v	v	$\mathcal{X}$. Namely, each vector v in $\mathbb{C}_{\text{fs}}^{\mathcal{X}}$ is by definition a function
rph8-g15q_F00425	21	■ 0.995	$\mathbb{C}_{\text{fs}}^{\mathcal{X}}$	$\mathbb{C}_{\text{fs}}^{\mathcal{X}}$	$\mathcal{X}$. Namely, each vector v in $\mathbb{C}_{\text{fs}}^{\mathcal{X}}$ is by definition a function
rph8-g15q_F00426	21	■ 1.000	$v:\mathcal{X}\rightarrow\mathbb{C}$	$v : \mathcal{X} \rightarrow \mathbb{C}$	$v : \mathcal{X} \rightarrow \mathbb{C}$ such that $v(x) = 0$ for all but finitely many $x \in \mathcal{X}$.
rph8-g15q_F00427	21	■ 1.000	$v(x)=0$	$v(x) = 0$	$v : \mathcal{X} \rightarrow \mathbb{C}$ such that $v(x) = 0$ for all but finitely many $x \in \mathcal{X}$.
rph8-g15q_F00428	21	■ 0.999	$v, w \in \mathbb{C}_{\text{fs}}^{\mathcal{X}}$	$v, w \in \mathbb{C}_{\text{fs}}^{\mathcal{X}}$	signs 0 to every $x \in \mathcal{X}$. The sum of two functions $v, w \in \mathbb{C}_{\text{fs}}^{\mathcal{X}}$
rph8-g15q_F00429	21	■ 1.000	$v+w$	$v + w$	is the function $v + w$ whose value on $x \in \mathcal{X}$ is $(v + w)(x) :=$
rph8-g15q_F00430	21	■ 1.000	$(v+w)(x):=$	$(v + w)(x) :=$	is the function $v + w$ whose value on $x \in \mathcal{X}$ is $(v + w)(x) :=$
rph8-g15q_F00431	21	■ 1.000	$v(x)+w(x)$	$v(x) + w(x)$	$v(x) + w(x)$. Meanwhile, if v is such a function and $\lambda \in \mathbb{C}$
rph8-g15q_F00432	21	■ 1.000	$\lambda \in \mathbb{C}$	$\lambda \in \mathbb{C}$	$v(x) + w(x)$. Meanwhile, if v is such a function and $\lambda \in \mathbb{C}$
rph8-g15q_F00433	21	■ 1.000	$\lambda \cdot v$	$\lambda \cdot v$	is a scalar, then $\lambda \cdot v$, written as λv for short, is the function
rph8-g15q_F00434	21	■ 1.000	λv	λv	is a scalar, then $\lambda \cdot v$, written as λv for short, is the function
rph8-g15q_F00435	21	■ 1.000	$\lambda v(x)$	$\lambda v(x)$	whose value on $x \in \mathcal{X}$ is $\lambda v(x)$.
rph8-g15q_F00436	21	■ 1.000	δ_x	δ_x	The elements $x \in \mathcal{X}$ define a basis δ_x for $\mathbb{C}_{\text{fs}}^{\mathcal{X}}$, where the
rph8-g15q_F00437	21	■ 0.998	$\delta_x(x')=\delta_{xx'}$	$\delta_x(x') = \delta_{xx'}$	function δ_x acts by $\delta_x(x') = \delta_{xx'}$ that is 1 if $x' = x$ and 0 if $x' \neq$
rph8-g15q_F00438	21	■ 0.998	$x'=x$	$x' = x$	function δ_x acts by $\delta_x(x') = \delta_{xx'}$ that is 1 if $x' = x$ and 0 if $x' \neq$
rph8-g15q_F00439	21	■ 0.998	$x'\neq$	$x' \neq$	function δ_x acts by $\delta_x(x') = \delta_{xx'}$ that is 1 if $x' = x$ and 0 if $x' \neq$
rph8-g15q_F00440	21	■ 1.000	$ x\rangle$	$ x\rangle$	could write $ x\rangle$ so that $\langle x' x\rangle = \delta_{xx'}$, but that might be consid-
rph8-g15q_F00441	21	■ 1.000	$\langle x' x\rangle=\delta_{xx'}$	$\langle x' x \rangle = \delta_{xx'}$	could write $ x\rangle$ so that $\langle x' x\rangle = \delta_{xx'}$, but that might be consid-
rph8-g15q_F00442	21	■ 1.000	$\mathcal{X}=\mathbb{Z}_2^n=$	$\mathcal{X} = \mathbb{Z}_2^n =$	familiar to the quantum information theorist. Let $\mathcal{X} = \mathbb{Z}_2^n =$
rph8-g15q_F00443	21	■ 1.000	$\{0,1\}^n$	$\{0, 1\}^n$	$\{0, 1\}^n$ be the set of all arrays $x = (x_0, x_1, \dots, x_{n-1})$ whose

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rph8-g15q_F00444	21	■ 1.000	$x=\left(x_{\{0\}},\ x_{\{1\}},\ \ldots,\ x_{\{n-1\}}\right)$	$x = (x_0, x_1, \dots, x_{n-1})$	$\{0, 1\}^n$ be the set of all arrays $x = (x_0, x_1, \dots, x_{n-1})$ whose
rph8-g15q_F00445	21	■ 1.000	$\left\{0,1,2,\ \ldots,\ 2^{\{n\}}-1\right\}$	$\{0, 1, 2, \dots, 2^n - 1\}$	representation of a number $\{0, 1, 2, \dots, 2^n - 1\}$. Then, the
rph8-g15q_F00446	21	■ 1.000	$\mathbb{C}^{\mathcal{X}}$	$\mathbb{C}^{\mathcal{X}}$	vector space $\mathbb{C}^{\mathcal{X}}$ generated by $\mathcal{X}$ is precisely (i.e., naturally
rph8-g15q_F00447	21	■ 1.000	$2^{\{n\}}$	2^n	is a finite set with 2^n elements.) From this perspective, each
rph8-g15q_F00448	21	■ 1.000	$L_{\{f\}}: \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow$	$L_f : \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow$	this function, define the linear transformation $L_f : \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow$
rph8-g15q_F00449	21	■ 0.995	$\mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$	$\mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$	$\mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$ as follows. By writing the basis elements of $\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$ and
rph8-g15q_F00450	21	■ 0.766	$ y\rangle$	$ y\rangle$	$\mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$ as $ x\rangle$ and $ y\rangle$, respectively, the linear transformation L_f
rph8-g15q_F00451	21	■ 0.766	$L_{\{f\}}$	L_f	$\mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$ as $ x\rangle$ and $ y\rangle$, respectively, the linear transformation L_f
rph8-g15q_F00452	21	■ 1.000	$ f(x)\rangle$	$ f(x)\rangle$	sends each basis vector $ x\rangle$ to $ f(x)\rangle$, which is another ba-
rph8-g15q_F00453	21	■ 1.000	$\mathbb{C}^{\mathcal{Y}}$	$\mathbb{C}^{\mathcal{Y}}$	sis vector in $\mathbb{C}^{\mathcal{Y}}$. Since a linear transformation is uniquely
rph8-g15q_F00454	21	■ 1.000	$L_{\{f\}}: \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$	$L_f : \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$	transforming $f : \mathcal{X} \rightarrow \mathcal{Y}$ to $L_f : \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$ is often used in
rph8-g15q_F00455	21	■ 0.871	$(V,\ 0,+,\ \cdot)$	$(V, 0, +, \cdot)$	going the other way that takes a vector space $(V, 0, +, \cdot)$ and
rph8-g15q_F00456	22	■ 1.000	$F(\mathcal{X})=$	$F(\mathcal{X}) =$	ing a set $\mathcal{X}$, constructing the associated vector space $F(\mathcal{X}) =$
rph8-g15q_F00457	22	■ 0.562	$G(F(\mathcal{X}))=\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$	$G(F(\mathcal{X})) = \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$	gives a <i>set</i> $G(F(\mathcal{X})) = \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$.
rph8-g15q_F00458	22	■ 1.000	$G \circ F$	$G \circ F$	hand, we have $G \circ F$, which we just described. On the other
rph8-g15q_F00459	22	■ 0.977	$\mathrm{id}_{\mathrm{Set}}$	$\mathrm{id}_{\mathrm{Set}}$	hand, we also have the functor $\mathrm{id}_{\mathrm{Set}} : \mathbf{Set} \rightarrow \mathbf{Set}$ that acts
rph8-g15q_F00460	22	■ 0.955	$\eta: \mathrm{id}_{\mathrm{Set}} \Rightarrow$	$\eta : \mathrm{id}_{\mathrm{Set}} \Rightarrow$	realize bit encoding as the natural transformation $\eta : \mathrm{id}_{\mathrm{Set}} \Rightarrow$
rph8-g15q_F00461	22	■ 0.877	$\eta_{\mathcal{X}}: \mathrm{id}_{\mathrm{Set}}(\mathcal{X}) \rightarrow G(F(\mathcal{X}))$	$\eta_{\mathcal{X}} : \mathrm{id}_{\mathrm{Set}}(\mathcal{X}) \rightarrow G(F(\mathcal{X}))$	$\eta_{\mathcal{X}} : \mathrm{id}_{\mathrm{Set}}(\mathcal{X}) \rightarrow G(F(\mathcal{X}))$, which is a function of the form
rph8-g15q_F00462	22	■ 1.000	$\eta_{\mathcal{X}}: \mathcal{X} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$	$\eta_{\mathcal{X}} : \mathcal{X} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$	$\eta_{\mathcal{X}} : \mathcal{X} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$. The definition of this function is precisely
rph8-g15q_F00463	22	■ 1.000	$\eta_{\mathcal{X}}(x)= x\rangle \in \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$	$\eta_{\mathcal{X}}(x) = x\rangle \in \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$	the vector $\eta_{\mathcal{X}}(x) = x\rangle \in \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$. So what does the naturality
rph8-g15q_F00464	22	■ 1.000	$\{ x\rangle\}$	$\{ x\rangle\}$	how it acts on the basis $\{ x\rangle\}$. Naturality of η then says that
rph8-g15q_F00465	22	■ 1.000	$\eta_{\mathcal{Y}}(f(x))=L_f(\eta_{\mathcal{X}}(x))$	$\eta_{\mathcal{Y}}(f(x)) = L_f(\eta_{\mathcal{X}}(x))$	$\eta_{\mathcal{Y}}(f(x)) = L_f(\eta_{\mathcal{X}}(x))$, which reads $ f(x)\rangle = L_f x\rangle$, a com-
rph8-g15q_F00466	22	■ 1.000	$ f(x)\rangle=L_f x\rangle$	$ f(x)\rangle = L_f x\rangle$	$\eta_{\mathcal{Y}}(f(x)) = L_f(\eta_{\mathcal{X}}(x))$, which reads $ f(x)\rangle = L_f x\rangle$, a com-
rph8-g15q_F00467	22	■ 1.000	$\mathbb{B}G$	$\mathbb{B}G$	group and let $\mathbb{B}G$ be the category consisting of only a sin-
rph8-g15q_F00468	22	■ 1.000	$\alpha: \mathbb{B}G \rightarrow$	$\alpha : \mathbb{B}G \rightarrow$	category. A G -set can then be viewed as a functor $\alpha : \mathbb{B}G \rightarrow$
rph8-g15q_F00469	22	■ 1.000	g^{-1}	g^{-1}	invertible, there exists a g^{-1} such that $gg^{-1} = 1_G = g^{-1}g$,
rph8-g15q_F00470	22	■ 1.000	$gg^{-1}=1_G=g^{-1}g$	$gg^{-1} = 1_G = g^{-1}g$	invertible, there exists a g^{-1} such that $gg^{-1} = 1_G = g^{-1}g$,
rph8-g15q_F00471	22	■ 0.987	$\mathrm{id}_{\mathcal{X}}=\alpha_e=\alpha_{gg^{-1}}=\alpha_g\circ\alpha_{g^{-1}}$	$\mathrm{id}_{\mathcal{X}} = \alpha_e = \alpha_{gg^{-1}} = \alpha_g \circ \alpha_{g^{-1}}$	of α , we have that $\mathrm{id}_{\mathcal{X}} = \alpha_e = \alpha_{gg^{-1}} = \alpha_g \circ \alpha_{g^{-1}}$, and simi-
rph8-g15q_F00472	22	■ 0.693	$\mathrm{id}_{\mathcal{X}}=\alpha_{g^{-1}}\circ\alpha_g$	$\mathrm{id}_{\mathcal{X}} = \alpha_{g^{-1}} \circ \alpha_g$	larly $\mathrm{id}_{\mathcal{X}} = \alpha_{g^{-1}} \circ \alpha_g$, which proves that α_g is a bijection with
rph8-g15q_F00473	22	■ 0.693	α_g	α_g	larly $\mathrm{id}_{\mathcal{X}} = \alpha_{g^{-1}} \circ \alpha_g$, which proves that α_g is a bijection with
rph8-g15q_F00474	22	■ 1.000	$\alpha_{g^{-1}}$	$\alpha_{g^{-1}}$	inverse $\alpha_{g^{-1}}$.
rph8-g15q_F00475	22	■ 0.986	α,β	α, β	Now, consider <i>two</i> such G -sets, viewed as functors $\alpha, \beta :$
rph8-g15q_F00476	22	■ 0.939	$\mathbb{B}G \rightarrow$	$\mathbb{B}G \rightarrow$	$\mathbb{B}G \rightarrow \mathbf{Set}$, where $\mathcal{X} := \alpha(\bullet)$ and $\mathcal{Y} := \beta(\bullet)$. A natural trans-
rph8-g15q_F00477	22	■ 0.939	$\mathcal{X}:=\alpha(\bullet)$	$\mathcal{X} := \alpha(\bullet)$	$\mathbb{B}G \rightarrow \mathbf{Set}$, where $\mathcal{X} := \alpha(\bullet)$ and $\mathcal{Y} := \beta(\bullet)$. A natural trans-
rph8-g15q_F00478	22	■ 0.939	$\mathcal{Y}:=\beta(\bullet)$	$\mathcal{Y} := \beta(\bullet)$	$\mathbb{B}G \rightarrow \mathbf{Set}$, where $\mathcal{X} := \alpha(\bullet)$ and $\mathcal{Y} := \beta(\bullet)$. A natural trans-
rph8-g15q_F00479	22	■ 0.949	$f: \alpha \Rightarrow \beta$	$f : \alpha \Rightarrow \beta$	formation $f : \alpha \Rightarrow \beta$ assigns to the only object $\bullet$ of $\mathbb{B}G$ a
rph8-g15q_F00480	22	■ 0.949	$\bullet$	$\bullet$	formation $f : \alpha \Rightarrow \beta$ assigns to the only object $\bullet$ of $\mathbb{B}G$ a
rph8-g15q_F00481	25	■ 1.000	w^*	w^*	uct measures to the tensor product of semifinite w^* -algebras,
rph8-g15q_F00482	25	■ 1.000	$x,\ y,\ \ldots \in \mathbb{R}$	$x, y, \dots \in \mathbb{R}$	whose objects are elements $x, y, \dots \in \mathbb{R}$ and where there is
rph8-g15q_F00483	25	■ 1.000	$x \rightarrow y$	$x \rightarrow y$	a morphism $x \rightarrow y$ whenever $x \leqslant y$. This is a category due
rph8-g15q_F00484	25	■ 1.000	$x \leqslant y$	$x \leqslant y$	a morphism $x \rightarrow y$ whenever $x \leqslant y$. This is a category due
rph8-g15q_F00485	25	■ 1.000	$\alpha \in \mathbb{R}^2$	$\alpha \in \mathbb{R}^2$	in category theory. The vector $\alpha \in \mathbb{R}^2$ is identified with the
rph8-g15q_F00486	25	■ 0.999	$\alpha: \mathbb{R} \rightarrow \mathbb{R}^2$	$\alpha : \mathbb{R} \rightarrow \mathbb{R}^2$	additive map $\alpha : \mathbb{R} \rightarrow \mathbb{R}^2$ sending $1 \in \mathbb{R}$ to α .

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rph8-g15q_F00487	25	■ 0.999	<code>1 \in \mathbb{R}</code>	$1 \in \mathbb{R}$	additive map $\alpha : \mathbb{R} \rightarrow \mathbb{R}^2$ sending $1 \in \mathbb{R}$ to α .
rph8-g15q_F00488	25	■ 1.000	<code>\operatorname{Tr}[\sigma\, 0]</code>	$\operatorname{Tr}[\sigma O]$	$\operatorname{Tr}[\sigma O]$ defines a hyperplane $\{\sigma \in \mathcal{S}(\mathcal{H}) : \operatorname{Tr}[\sigma O] = 0\}$ in
rph8-g15q_F00489	25	■ 1.000	<code>\{\sigma \in \mathscr{S}(\mathcal{H}) : \operatorname{Tr}[\sigma\, 0]=0\}</code>	$\{\sigma \in \mathcal{S}(\mathcal{H}) : \operatorname{Tr}[\sigma O] = 0\}$	$\operatorname{Tr}[\sigma O]$ defines a hyperplane $\{\sigma \in \mathcal{S}(\mathcal{H}) : \operatorname{Tr}[\sigma O] = 0\}$ in